1 Setting

Hypotheses are axiom systems produced by programs; data are sentences labelled true or false, and a hypothesis is compatible with the data when it is consistent and proves every datum. Randomness enters as in the one-tape version of Solomonoff operator induction in the companion paper (Hänni and Claude (Anthropic), 2026b). Two further works are cited by name: the companion paper on steeper priors (Hänni and Claude (Anthropic), 2026c) and our earlier report (Hänni and Claude (Anthropic), 2026a). Proofs for this section and the next are in Appendix A.

1.1 Language, background and data

Definition 1.1 (Setting and data)**.** $\mathcal{L}$ is a countable first-order language with equality and computable syntax, $\mathrm{Sent} := \mathrm{Sent}_{\mathcal{L}}$, and $\vdash$ is derivability in a fixed sound and complete calculus. The *background* $B \subseteq \mathrm{Sent}$ is decidable and consistent; by default $B = \emptyset$. Put $\mathrm{Cn}(\Gamma) := \{\varphi : B \cup \Gamma \vdash \varphi\}$. "$\Gamma$ proves φ" means $B \cup \Gamma \vdash \varphi$; Γ is *consistent* if $B \cup \Gamma$ is; Γ *decides* φ if it proves φ or $\neg\varphi$; $\varphi^1 := \varphi$ and $\varphi^0 := \neg\varphi$.

A *data set* is a finite partial function $D : \mathrm{Sent} \rightharpoonup \{0,1\}$ (label 1 means true); a *three-valued* data set maps into $Y_3 := \{1, 0, \iota\}$, where ι means "independent". Put $D^{\pm} := \{\varphi^{D(\varphi)} : D(\varphi) \in \{0,1\}\}$, and let ψ_D be the conjunction of $D^{\pm}$ in a fixed canonical order ($\top$ if $D^{\pm} = \emptyset$); D is *consistent* if $B \cup D^{\pm}$ is. $D + (\varphi, y)$ is defined for $\varphi \notin \mathrm{dom}\, D$. A *reveal sequence* $(\varphi_t, y_t)_{t \le n}$ is consistent if $D_n := \{(\varphi_t, y_t) : t \le n\}$ is a partial function; D_t are its prefixes. An infinite stream D_∞ has $D_\infty^{\pm} := \bigcup_n D_n^{\pm}$. For a map ν from data sets to $[0,1]$ the *conditional* is $\nu(y \mid D, \varphi) := \nu(D + (\varphi, y))/\nu(D)$ for $\varphi \notin \mathrm{dom}\, D$ (0 if $\nu(D) = 0$), and $\nu(y \mid D, \varphi) := \mathbf{1}[y = D(\varphi)]$ for $\varphi \in \mathrm{dom}\, D$ (the repeated-input convention of the companion paper).

An r.e. background would serve as well, since by Craig's trick it has a decidable axiomatisation with the same consequences.

Algorithmic notation. V is a fixed universal prefix machine, K prefix complexity, $\mathbf{m}(i) := \sum_{V(p)=i} 2^{-|p|} \ge 2^{-K(i)}$ and $\Omega_V := \sum_{p \in \mathrm{dom}\, V} 2^{-|p|}$; W_e is the e-th r.e. set of sentences. Put $\Omega_V^{\ge \ell} := \sum\{2^{-|p|} : p \in \mathrm{dom}\, V,\ |p| \ge \ell\}$, the tail of Ω_V, which tends to 0 as $\ell \to \infty$; and $K^{\mathrm{pad}}(n) := \min_{n' \ge n} K(n')$, which is nondecreasing and unbounded. $\mathbb{U}$ is the fair-coin measure on tapes $\omega \in \{0,1\}^\infty$, $[w]$ the cylinder of a finite string w, and $\omega_{1:n}$ the first n bits of ω. Logarithms are binary and losses are in bits. The constants $c, c', \ldots$ depend only on the fixed universal machines, the coding of syntax and B; "lsc" means lower semicomputable. We use two standard facts (see Li and Vitányi, 2008): $\mathbf{m}$ dominates every lsc discrete semimeasure up to a constant factor; and the conditional coding theorem: if $P(a \mid z)$ is lsc uniformly in (a, z) and $\sum_a P(a \mid z) \le 1$, then $K(a \mid z) \le -\log P(a \mid z) + K(P) + c$.

Definition 1.2 (Literal setting; fresh atoms)**.** In the *literal setting*, $\mathcal{L}$ contains a unary relation symbol R that does not occur in B, and closed terms t_w ($w \in \{0,1\}^*$) that some model of B interprets as pairwise distinct elements. *Fresh atoms* are sentences $R(t_w)$ for pairwise distinct words w; so $R(t_w)$ and $\neg R(t_w)$ are never two fresh atoms. The standard instance is the *literal language*: a constant $\mathtt{e}$, unary function symbols $\mathtt{s}_0, \mathtt{s}_1$, and the unary relation symbol R, with $t_w := \mathtt{s}_{w_1}(\cdots \mathtt{s}_{w_k}(\mathtt{e}) \cdots)$ and $\varphi_w := R(t_w)$, of $|w| + 2$ symbols; its *named structure* $\mathfrak{T}$ is the free term algebra on $\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1$. For a set of words $X \subseteq \{0,1\}^*$, D^X lists $(\varphi_w, \mathbf{1}[w \in X])$ in length-lexicographic order and f_X is the literal labeller of X. The *fit class* of an inducer is the set of X on whose sequence D^X it has bounded loss.

Every result stated for fresh atoms or on literal data assumes this setting. Two facts carry all such uses (Lemma A.14): a set of literals of R with at most one literal per word is consistent with B, and a consistent such set proves exactly its own members among the literals.

1.2 Theory machines and their laws

A hypothesis of Solomonoff axiom induction is produced by a machine that reads random bits and writes sentences. A function machine of Solomonoff operator induction must give each input at most one label; a theory machine may write anything, and consistency is a property of its output, a random event.

Definition 1.3 (Theory machine)**.** A *theory machine* is a c.e. set $G \subseteq \{0,1\}^* \times \mathrm{Sent}$ ("after reading w from the tape, emit χ"). For finite w put $A^G_w := \{\chi : (v,\chi) \in G,\ v \sqsubseteq w\}$, and for a tape ω let $A^G_\omega := \bigcup_n A^G_{\omega_{1:n}}$ be the axioms emitted on ω. Let

$$O^G_\chi := \{\omega : B \cup A^G_\omega \vdash \chi\}, \qquad \mathrm{Cons}^G := \{0,1\}^\infty \setminus O^G_\bot, \qquad E^G_D := O^G_{\psi_D} \cap \mathrm{Cons}^G.$$

The *law* of G is $\nu_G(D) := \mathbb{U}(E^G_D)$, the probability that the random theory is consistent and proves every datum. The *three-valued law* is $\nu^3_G(D) := \mathbb{U}(E^{3,G}_D)$ with $E^{3,G}_D := E^G_D \cap \bigcap_{D(\varphi)=\iota} (\{0,1\}^\infty \setminus (O^G_\varphi \cup O^G_{\neg\varphi}))$: in addition, the theory decides no ι-labelled sentence. On Cons^G the *trichotomy function* $\mathrm{tri}^G_\omega : \mathrm{Sent} \to Y_3$ is 1 at the sentences $B \cup A^G_\omega$ proves, 0 at those it refutes and ι elsewhere. (G_i) is a fixed effective enumeration of the c.e. subsets of $\{0,1\}^* \times \mathrm{Sent}$; $K(G) := \min\{K(i) : G_i = G\}$ and, for a law ν, $K(\nu) := \min\{K(i) : \nu_{G_i} = \nu\}$.

These events are measurable: O^G_χ is effectively open uniformly in χ and an index of G, Cons^G is a Π^0_1 class, and $\nu_G(D) = \mathbb{U}(O^G_{\psi_D}) - \mathbb{U}(O^G_\bot)$ (Lemma A.1); examples are in Appendix A.1. A theory machine can complete a law at random inside what the law allows; such randomised theories are how approximate laws enter Solomonoff axiom induction (**??**).

Definition 1.4 (Solomonoff axiom induction)**.** The *universal theory machine* is $G_U := \{(pw,\chi) : p \in \mathrm{dom}\, V,\ (w,\chi) \in G_{V(p)}\}$: it reads a self-delimiting program and runs that machine on the rest of the tape. Write $A_\omega := A^{G_U}_\omega$, $O_\chi := O^{G_U}_\chi$, $\mathrm{Cons} := \mathrm{Cons}^{G_U}$, $E_D := E^{G_U}_D$, $E^3_D := E^{3,G_U}_D$ and $\mathrm{tri}_\omega := \mathrm{tri}^{G_U}_\omega$.

- *SAI* is the one-tape proof-requiring law $M_A := \nu_{G_U}$. Put $M^3_A := \nu^3_{G_U}$, $L_A := -\log M_A$ and $\mathrm{Inc}_0 := \mathbb{U}(O_\bot)$, the probability that the random theory is inconsistent.
- The *posterior* after D with $M_A(D) > 0$ is the law of A_ω under $\mathbb{U}(\cdot \mid E_D)$; three-valued, under $\mathbb{U}(\cdot \mid E^3_D)$.
- *Trichotomy.* $\mathrm{Bel}_D(\chi) := \mathbb{U}(E_D \cap O_\chi)/\mathbb{U}(E_D)$ is the posterior probability that the random theory proves χ; $\mathrm{Dis}_D(\chi) := \mathrm{Bel}_D(\neg\chi)$; $\mathrm{Indep}_D(\chi) := 1 - \mathrm{Bel}_D(\chi) - \mathrm{Dis}_D(\chi)$.
- The *prior probability of provability* $\mathrm{Bel}_0(\chi) := M_A(\{(\chi,1)\}) = \mathbb{U}(O_\chi) - \mathrm{Inc}_0$ is not normalised: $\mathrm{Bel}_0(\top) = 1 - \mathrm{Inc}_0$. Its normalisation is $\widehat{\mathrm{Bel}}_0 := \mathrm{Bel}_0/(1 - \mathrm{Inc}_0) = \mathrm{Bel}_\emptyset$. The *plausibility* is $\mathrm{Pl}_0(\chi) := \mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\neg\chi)$, and $\mathrm{Pl}_D(\chi) := 1 - \mathrm{Bel}_D(\neg\chi)$.
- *Class probabilities.* For a class $\mathcal{C}$ of deductively closed theories, $\mathbb{U}_A(\mathcal{C}; D) := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent},\ \vdash \psi_D,\ \mathrm{Cn}(A_\omega) \in \mathcal{C}\}$ and $\mathbb{U}_A(\mathcal{C}) := \mathbb{U}_A(\mathcal{C}; \emptyset)$. For a theory T, $\mathbb{U}_A[T] := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent},\ \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T)\}$ and $\mathbb{U}_A(\mathrm{Ext}T) := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent},\ \mathrm{Cn}(A_\omega) \supseteq \mathrm{Cn}(T)\}$.

If $\omega = p\omega'$ with $p \in \operatorname{dom} V$, then $A_\omega = A_{\omega'}^{G_{V(p)}}$, so G_U reads one tape in two parts, a program and that program's coins; a tape with no prefix in $\operatorname{dom} V$ emits nothing. By **??**, $M_A(D) = \mathrm{Bel}_0(\psi_D)$, $M_A(b \mid D, \varphi) = \mathrm{Bel}_D(\varphi^b)$ and $M_A^3(\iota \mid D, \varphi) = \mathrm{Indep}_D(\varphi)$ for $\varphi \notin \operatorname{dom} D$: the predictions of SAI are posterior probabilities that the random theory proves, refutes, or leaves open the queried sentence.

1.3 Conventions and the two defects

Definition 1.5 (Loss conventions)**.** • *Semimeasure convention* (primary): the cumulative loss of a consistent reveal sequence is $\sum_{t \le n} -\log M_A(y_t \mid D_{t-1}, \varphi_t)$, which equals $L_A(D_n) - L_A(\emptyset)$ (**??**(d)); for the coin-free version it is $-\log W_A(D_n) + \log W_A(\emptyset)$.

- *Normalised convention*: each conditional is divided by $\mathrm{Bel}_D(\varphi) + \mathrm{Dis}_D(\varphi)$.
- *Three-valued convention*: labels in Y_3, scored by M_A^3, whose conditionals sum to 1 (**??**(f)).
- The consistency-requiring predictor (Definition 1.9) has its own loss, $\sum_t -\log \mathrm{Bel}^{\mathrm{nc}}_{D_{t-1}}(\varphi_t^{b_t})$.

Upper bounds proved in the semimeasure convention hold in the normalised convention, since each normalised conditional is at least the raw one. Lower bounds do not transfer: a lower bound in the semimeasure convention can fail in the normalised convention by an unbounded amount (**??**), and the normalised loss depends on the reveal order (Theorem A.33). Every lower bound below names its convention; statements about K_A, K_{fin} and the posterior need none.

The missing mass of M_A has two sources, and we keep both rather than renormalise.[1]

Remark 1.6 (Two defects). M_A has a *global* defect $1 - M_A(\emptyset) = \mathrm{Inc}_0 > 0$ and a *local* defect $\mathrm{Indep}_D(\varphi)$ at each query (**??**(b), (e)). The one-tape law M_F of SOI has only the local one ($M_F(\emptyset) = 1$), because function machines are consistent by definition. (The correspondence of the outcomes with those of function machines is in Appendix A.1.) Removing the independent mass and renormalising, or splitting it equally between the two labels, gives incoherent predictions (Examples 2.15, 2.16 and Proposition 2.17 of the earlier report); keeping it, there is a probability on the complete consistent extensions of B that lies between Bel_D and Pl_D (**??**(c)).

1.4 Versions and complexities of theories

Definition 1.7 (Versions)**.** As for SOI, the one-tape law has a coin-free version, a mixture over deterministic programs, and a Kolmogorov version, the shortest compatible program. For a data set D:

- the coin-free version, over enumerators: $W_A(D) := \sum_{p \in \operatorname{dom} V} 2^{-|p|} \mathbf{1}[B \cup \mathsf{W}_{V(p)} \text{ consistent and } \vdash \psi_D]$, which equals $\sum_e \mathbf{m}(e) \mathbf{1}[B \cup \mathsf{W}_e \text{ consistent and } \vdash \psi_D]$;
- $K_A(D) := \min\{|p| : p \in \operatorname{dom} V,\ B \cup \mathsf{W}_{V(p)} \text{ consistent and } \vdash \psi_D\}$ (∞ if there is none; the Kolmogorov version);
- W_A^{dec}, K_A^{dec}: the same over *deciders*, the $p \in \operatorname{dom} V$ such that $V(p)$ is an index of a total $\{0,1\}$-valued function d, read as the axiom set $\{\chi : d(\chi) = 1\}$ (Definition 6.2 of the earlier report);

[1] The proof-requiring and consistency-requiring variants, the three outcomes true, false and independent, and the choice to keep the defect of the semimeasure rather than renormalise first appeared in Hänni's working notes (Hänni, 2026).

- $W_A^{\mathrm{gen},t}$: the same over enumerators of the delay form (Definition E.10 of the earlier report), which write each axiom a within $O(t(|a|))$ steps of the previous one, and $K_A^{\mathrm{gen},t}(D)$, the least length of such an enumerator compatible with D;
- $K_{\mathrm{fin}}(D) := \min\{K(\psi) : \psi \in \mathrm{Sent},\ B \cup \{\psi\} \text{ consistent},\ B \cup \{\psi\} \vdash \psi_D\}$ and $W_{\mathrm{fin}}(D) := \sum_\psi \mathbf{m}(\psi)\mathbf{1}[B \cup \{\psi\} \text{ consistent},\ B \cup \{\psi\} \vdash \psi_D]$ (finite axiomatisations).

Deciders and enumerators give the same weights up to a constant factor, polynomial-delay enumerators too under the self-simulation assumption of the earlier report (**??**), and $K_A \le K_{\mathrm{fin}} + c \le K(\psi_D) + c$ on consistent data, where $K_{\mathrm{fin}} - K_A$ is unbounded (Remark A.2 and **??**).

Definition 1.8 (Complexities of theories)**.** For an r.e. theory T: $[T] := \{e : \mathrm{Cn}(\mathsf{W}_e) = \mathrm{Cn}(T)\}$; $K_\equiv(T) := \min_{e\in[T]} K(e)$, the complexity of T up to deductive equivalence; $\mathbf{m}_\equiv(T) := \sum_{e\in[T]} \mathbf{m}(e)$; $\mathrm{Ext}(T) := \{e : B \cup \mathsf{W}_e \text{ consistent},\ \mathrm{Cn}(\mathsf{W}_e) \supseteq \mathrm{Cn}(T)\}$ and $\mathbf{m}(\mathrm{Ext}T) := \sum_{e\in \mathrm{Ext}T} \mathbf{m}(e)$ (their relations are in Remark A.3).

1.5 Variants and questions

A weaker requirement than proof is consistency with the data; **??** explains why we take the proof-requiring law as primary.

Definition 1.9 (Variants)**.** Let $\varepsilon_\emptyset := \mathbb{U}\{\omega : A_\omega = \emptyset\}$, the mass of the empty theory; $\varepsilon_\emptyset \ge 2^{-|p_\emptyset|} > 0$ for a V-program $p_\emptyset$ of an index of the empty machine. The *consistency-requiring* 0/1 weight is $M_A^{\mathrm{nc}}(D) := \mathbb{U}\{\omega : B \cup A_\omega \cup D^\pm \text{ consistent}\}$. Its posterior is the law of A_ω given that event, and its predictor uses the hypothesis together with the data: $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) := \mathbb{U}\{\omega : B \cup A_\omega \cup D^\pm \text{ consistent and } \vdash \varphi\}/M_A^{\mathrm{nc}}(D)$ and $\mathrm{Dis}_D^{\mathrm{nc}}(\varphi) := \mathrm{Bel}_D^{\mathrm{nc}}(\neg\varphi)$.

A graded consistency-requiring weight $M_A^{1/2}$ and a generative law Ξ of unlabelled data are defined in Definitions A.20 and A.36.

Definition 1.10 (Questions)**.** A *question* is a finite set $\mathcal{Q} = \{\varphi_1, \ldots, \varphi_k\}$ of pairwise B-incompatible sentences whose disjunction B proves, its *direct answers*; a yes/no question is $\{\varphi, \neg\varphi\}$. A theory *answers* $\mathcal{Q}$ if it proves some φ_i. A *conditional question* is a pair (α, ℓ), read "given α, is ℓ the case?". Under the *encoding* (E3), a theory T is compatible with the conditional datum (α, ℓ) iff $B \cup T \cup \{\alpha\}$ is consistent and proves ℓ; equivalently, $T \vdash \alpha \to \ell$ and $T \nvdash \neg\alpha$.

This follows the tradition of erotetic logic (cf. Belnap and Steel, 1976); results that need only pairwise incompatibility say so. (E3) makes "given α" hypothetical: a compatible theory may not refute an antecedent, so no single world fits question data with two incompatible antecedents (more in Appendix A.1).

A Proofs for the setting and the basic theorems

This appendix proves the results of Section 1 and **??** in their order, and states and proves the results that the main text only mentions. Throughout, G is a theory machine and the notation is that of Definitions 1.3 and 1.4.

A.1 The setting

Lemma A.1 (Measurability and the basic form)**.** *(a)* $O^G_\chi = \bigcup\{[w] : B \cup A^G_w \vdash \chi\}$ *is an effectively open class, uniformly in an index of* G *and in* χ*; hence* $\mathbb{U}(O^G_\chi)$ *is left-c.e., uniformly. (b)* Cons^G *is a* Π^0_1 *class, and* $O^G_\perp \subseteq O^G_\chi$ *for every* χ*. (c)* $\nu_G(D) = \mathbb{U}(O^G_{\psi_D}) - \mathbb{U}(O^G_\perp)$*, and* $E^{3,G}_D$ *is Borel.*

Proof. (a) If $B \cup A^G_\omega \vdash \chi$, a derivation uses finitely many axioms $\chi_1, \dots, \chi_r \in A^G_\omega$, each emitted under some prefix of ω. The prefixes of ω are linearly ordered, so the longest of them, w, has $\chi_1, \dots, \chi_r \in A^G_w$, and every $\omega' \sqsupseteq w$ has $B \cup A^G_{\omega'} \vdash \chi$. Conversely $[w] \subseteq O^G_\chi$ whenever $B \cup A^G_w \vdash \chi$. The set of pairs (w, χ) with $B \cup A^G_w \vdash \chi$ is c.e.: enumerate G and all derivations, using that B is decidable. (b) Cons^G is the complement of an effectively open class, and an inconsistent set proves everything. (c) $E^G_D = O^G_{\psi_D} \setminus O^G_\perp$ with $O^G_\perp \subseteq O^G_{\psi_D}$; and $E^{3,G}_D$ is E^G_D intersected with finitely many closed classes. □

Examples of theory machines. (i) A deterministic r.e. theory W_e: $G = \{(\epsilon, \chi) : \chi \in \mathsf{W}_e\}$, with $\nu_G(D) = \mathbf{1}[B \cup \mathsf{W}_e$ consistent and $\vdash \psi_D]$. (ii) Fair-coin facts on a family $(\alpha_j)_{j \geq 1}$ of sentences: $G = \{(w, \alpha_j^{w_j}) : |w| = j\}$. (iii) A theory plus coin facts: the union of (i) and (ii). (iv) A finite mixture of finite theories $T_1, \dots, T_r$: $G = \{(w_j, \chi) : \chi \in T_j\}$ for a prefix-free set $\{w_1, \dots, w_r\}$, which gives T_j weight $2^{-|w_j|}$; a tape with no prefix among the w_j emits nothing, and its theory is the empty one, consistent since B is.

Remark A.2 (Deciders, enumerators and polynomial-delay enumerators agree)*.* There is c such that for every D: (a) $2^{-c} W^{\mathrm{dec}}_A(D) \leq W_A(D) \leq 2^c W^{\mathrm{dec}}_A(D)$ and $|K_A(D) - K^{\mathrm{dec}}_A(D)| \leq c$; (b) under **??**, for t time-constructible, nondecreasing and superadditive with $t(m) \geq m$ and $t(m) \geq m^{a^*}$ (a^* fixed by the universal machine): $2^{-c} W_A(D) \leq W^{\mathrm{gen},t}_A(D) \leq 2^c W_A(D)$, and $|K_A(D) - K^{\mathrm{gen},t}_A(D)| \leq c$; (c) $K_A(D) \leq K_{\mathrm{fin}}(D) + c \leq K(\psi_D) + c$ if D is consistent; if D is inconsistent, $K_A(D) = K_{\mathrm{fin}}(D) = \infty$, while $K(\psi_D)$ is finite.

Proof. (a) *Deciders to enumerators.* By universality of V there is a string r_1 such that, whenever p is a decider, $V(r_1 p)$ is an index of the r.e. set $\{\chi : \{V(p)\}(\chi) = 1\}$. The map $p \mapsto r_1 p$ is injective, adds $|r_1|$ to the length and preserves the axiom set. *Enumerators to deciders (Craig's trick).* Write $\chi^{\wedge 1} := \chi$ and $\chi^{\wedge(m+1)} := \chi \wedge \chi^{\wedge m}$. For an index e fix the step-by-step enumeration of W_e and let $A^{\mathrm{C}}_e := \{\chi^{\wedge(s+1)} : \chi$ is enumerated at step $s\}$. A sentence θ has at most two readings $\theta = \chi^{\wedge m}$: $(\chi, m) = (\theta, 1)$, and, if θ is a conjunction, χ its left conjunct with m the number of times χ can be stripped off. (If $\theta = \chi^{\wedge m}$ with $m \geq 2$, then χ is the left conjunct of θ, and stripping stops exactly at χ, because χ is not of the form $\chi \wedge X$.) So A^{C}_e is decided by the procedure: find the readings, and for each reading (χ, m) run the enumeration for exactly $m - 1$ steps and check whether χ appears at step $m - 1$. It halts on every input, also when W_e is finite. Since $\chi^{\wedge m}$ is logically equivalent to χ, $\mathrm{Cn}(A^{\mathrm{C}}_e) = \mathrm{Cn}(\mathsf{W}_e)$. A fixed string r_2 makes $V(r_2 p)$ an index of this decider for $e = V(p)$; the map $p \mapsto r_2 p$ is injective and adds $|r_2|$. Both maps preserve consistency and what is proved, so summing over programs, and taking shortest programs, gives (a) with $c := \max(|r_1|, |r_2|)$. This is the argument of Theorem 6.3 of our earlier report (Hänni and Claude (Anthropic), 2026a), here between two kinds of axiom hypotheses rather than between axiom and labeller hypotheses.

(b) Under **??**, Theorem 6.14(a) of the earlier report shows that $W^{\mathrm{gen},t}_A$ is within a constant factor of the coin-free consistent mixture W^{cons} of labellers, and its Theorem 6.3 shows $W^{\mathrm{cons}} \asymp W^{\mathrm{dec}}_A$; combined with (a) this gives the claim for $W^{\mathrm{gen},t}_A$. For the Kolmogorov versions, $|K_A(D) - K^{\mathrm{gen},t}_A(D)| \leq c$, we use the map behind it. An enumerator p of A is sent to the padded enumerator

of the semi-decider of A ("yes on χ iff χ is enumerated") of Lemma E.12 of the earlier report; that lemma, proved there under **??**, gives an enumerator of the delay form with the same consequences as A, injectively and with length at most $|p| + c$. Conversely, an enumerator of the delay form is an enumerator. The translation between the universal machine of the earlier report and programs of V that print indices costs a constant. We do not re-derive **??**.

(c) If ψ witnesses $K_{\text{fin}}(D)$, a fixed prefix r_3 turns a shortest program for ψ into a program for an index of $\{\psi\}$, which is consistent with B and proves ψ_D; so $K_A \leq K_{\text{fin}} + |r_3|$. For consistent D, ψ_D itself is a candidate for K_{fin}. If D is inconsistent, no consistent set proves ψ_D. □

Remark A.3 (Complexities of consistent theories). If $B \cup T$ is consistent, then $[T] \subseteq \text{Ext}T$, so $2^{-K_{\equiv}(T)} \leq \mathbf{m}_{\equiv}(T) \leq \mathbf{m}(\text{Ext}T)$, and $\mathbb{U}_A[T] \geq 2^{-c}\mathbf{m}_{\equiv}(T)$, $\mathbb{U}_A(\text{Ext}T) \geq 2^{-c}\mathbf{m}(\text{Ext}T)$ (**??**(i)). The hypothesis is needed: for $T = \{\bot\}$, $\mathbb{U}_A[T] = 0$ and $\text{Ext}T = \emptyset$, while $\mathbf{m}_{\equiv}(T) \geq 2^{-K_{\equiv}(T)} > 0$.

The two defects and function machines. The outcomes of a theory machine at a sentence φ correspond to those of a function machine at an input: "proves φ" is a Σ_1 property of the tape, like "halts with output 1"; "independent" is Π_1, like "diverges"; "inconsistent" corresponds to "halts with both outputs", which a function machine cannot do.

The encoding (E3). (E3) is not a single datum of M_A^3: it combines "the label of $\alpha \to \ell$ is 1" with "the label of $\neg\alpha$ is not 1", a union of two M_A^3-events. The two-valued encoding by the datum $(\alpha \to \ell, 1)$ is degenerate, since a single world satisfying the law fits all of its question data (**??**); it is never used for conditional data.

A.2 Structure and updating

Proof of **??**. (a) If D is inconsistent, every set proving ψ_D is inconsistent, so $E_D^G = \emptyset$. If D is consistent, the deterministic machine $G_D := \{(\epsilon, \psi_D)\}$ has $\nu_{G_D}(D) = 1$, so $M_A(D) \geq 2^{-K(G_D)} > 0$ by **??**. Three-valued: $B \cup \{\psi_D\}$ decides exactly what $B \cup D^{\pm}$ decides, so if $B \cup D^{\pm}$ decides no ι-labelled sentence then $\nu^3_{G_D}(D) = 1$; conversely, if $B \cup D^{\pm}$ decides an ι-labelled sentence, so does every set proving ψ_D.

(b) O_χ^G depends only on the B-equivalence class of χ, and $\psi_{D+(\varphi,b)}$ is equivalent to $\psi_D \wedge \varphi^b$. On E_D^G the random theory is consistent, so exactly one of three things happens: it proves φ, it proves $\neg\varphi$, or it decides neither. The three events are $E^G_{D+(\varphi,1)}$, $E^G_{D+(\varphi,0)}$ and $E^{3,G}_{D+(\varphi,\iota)}$. For G_U, dividing by $\mathbb{U}(E_D)$ shows that the third term is $M_A(D)(1 - \text{Bel}_D(\varphi) - \text{Dis}_D(\varphi))$.

(c) $B \cup A \vdash \psi_D$ iff $\text{Cn}(D^{\pm}) \subseteq \text{Cn}(A)$. If $B \cup D^{\pm} \vdash \varphi^b$, then $\text{Cn}(D^{\pm} \cup \{\varphi^b\}) = \text{Cn}(D^{\pm})$, and $D + (\varphi, 1-b)$ is inconsistent, so (a) applies to it.

(d) At a step with $\varphi_t \notin \text{dom}\, D_{t-1}$ the factor is $M_A(D_t)/M_A(D_{t-1})$; at a repeat (necessarily with the stored label, since the sequence is consistent) it is 1 and $D_t = D_{t-1}$. The product telescopes to $M_A(D_n)/M_A(\emptyset)$. If some $M_A(D_t) = 0$, the first such step has factor 0, and both sides are $+\infty$ by the monotonicity in (b). Dependence only on $\text{Cn}(D_n^{\pm})$ is (c).

(e) $M_A(\emptyset) = \mathbb{U}(\text{Cons})$. The tapes beginning with a V-program for an index of the empty machine have $A_\omega = \emptyset$, consistent since B is; the tapes beginning with a program for the machine $\{(\epsilon, \bot)\}$ are inconsistent. Both sets have positive measure.

(f) As (b), with $E_D^{3,G}$ in place of E_D^G; and $\nu_G^3(\emptyset) = \mathbb{U}(\text{Cons}^G) = \nu_G(\emptyset)$.

(g) By definition $E_D^{3,G} = \{\omega \in \text{Cons}^G : \text{tri}_\omega^G \supseteq D\}$, and for two-valued D this is E_D^G. □

Proof of **??**. (a) A theory proves ψ_D and χ iff it proves $\psi_D \wedge \chi$, so $E_D \cap O_\chi = O_{\psi_D \wedge \chi} \setminus O_\bot$, and both sides equal $\mathbb{U}(O_{\psi_D \wedge \chi} \setminus O_\bot)/\mathbb{U}(O_{\psi_D} \setminus O_\bot)$. The normalising constant $1 - \text{Inc}_0$ cancels.

(b) Let $P := \mathbb{U}(\cdot \mid E_D)$ and $Q_i := O_{\chi_i}$. Then $\bigcap_{i\in I} Q_i = O_{\bigwedge_{i\in I}\chi_i}$, since on one tape a theory proves each χ_i iff it proves their conjunction, and $\bigcup_i Q_i \subseteq O_{\bigvee_i \chi_i}$. Inclusion–exclusion for $P(\bigcup_i Q_i)$ gives the inequality. Bel_0 is the unnormalised case, with P the restriction of $\mathbb{U}$ to Cons.

(c) Fix an enumeration (σ_k) of Sent. For $\omega \in \mathrm{Cons}$ let $T_0(\omega) := B \cup A_\omega$, and $T_{k+1} := T_k \cup \{\sigma_k\}$ if that set is consistent, $T_{k+1} := T_k \cup \{\neg\sigma_k\}$ otherwise. $T_\infty(\omega) := \bigcup_k T_k$ is a complete consistent extension of B: it is consistent by compactness, and complete and deductively closed because it is maximal consistent. Each choice depends on whether $B \cup A_\omega \cup F$ is consistent for a finite set F, that is, on membership of ω in the open class $O_{\neg\bigwedge F}$; so $\omega \mapsto T_\infty(\omega)$ is Borel. Let P_D be its law under $\mathbb{U}(\cdot \mid E_D)$. If $B \cup A_\omega \vdash \chi$ then $\chi \in T_\infty$, and if $\chi \in T_\infty$ then $B \cup A_\omega \nvdash \neg\chi$. So $\mathrm{Bel}_D \le P_D \le \mathrm{Pl}_D$. □

Proposition A.4 (Consistency-requiring updating is Dempster conditioning)**.** *Let $\psi := \psi_D$ and $M_A^{\mathrm{nc}}(D) > 0$. Then $M_A^{\mathrm{nc}}(D) = \mathrm{Pl}_0(\psi)$ and $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) = \dfrac{\mathrm{Bel}_0(\varphi \vee \neg\psi) - \mathrm{Bel}_0(\neg\psi)}{\mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\neg\psi)} = \dfrac{\widehat{\mathrm{Bel}}_0(\varphi \vee \neg\psi) - \widehat{\mathrm{Bel}}_0(\neg\psi)}{1 - \widehat{\mathrm{Bel}}_0(\neg\psi)}$.*

Proof. For consistent $B \cup A$: $B \cup A \cup \{\psi\}$ is consistent iff $B \cup A \nvdash \neg\psi$, and $B \cup A \cup \{\psi\} \vdash \varphi$ iff $B \cup A \vdash \varphi \vee \neg\psi$ (the deduction theorem; ψ is a sentence). So $M_A^{\mathrm{nc}}(D) = \mathbb{U}(\mathrm{Cons}) - \mathbb{U}(\mathrm{Cons} \cap O_{\neg\psi}) = \mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\neg\psi) = \mathrm{Pl}_0(\psi)$. Since $B \cup A \vdash \neg\psi$ implies $B \cup A \vdash \varphi \vee \neg\psi$, the numerator of $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi)$ is $\mathbb{U}(\mathrm{Cons} \cap O_{\varphi\vee\neg\psi}) - \mathbb{U}(\mathrm{Cons} \cap O_{\neg\psi}) = \mathrm{Bel}_0(\varphi \vee \neg\psi) - \mathrm{Bel}_0(\neg\psi)$. Dividing numerator and denominator by $1 - \mathrm{Inc}_0$ gives the form with $\widehat{\mathrm{Bel}}_0$. □

A.3 Computability

Proposition A.5. *$M_A(D) = \mathbb{U}(O_{\psi_D}) - \mathrm{Inc}_0$, where $\mathbb{U}(O_{\psi_D})$ is left-c.e. uniformly in D and $\mathrm{Inc}_0 = \mathbb{U}(O_\perp)$ is left-c.e.*

Proof. Lemma A.1(a), (c) for $G = G_U$. □

The following lemma is a form of a theorem of Demuth (Demuth, 1975), as it is attributed in the literature: a left-c.e. real plus a random left-c.e. real is random (the converse is attributed to Downey et al., 2002). We prove the form we need, for completeness. For a real $x \in (0,1)$, $x_{1:n}$ denotes its first n binary digits; we use the Levin–Schnorr characterisation (standard; see Li and Vitányi, 2008): x is Martin-Löf random iff $K(x_{1:n}) \ge n - c_x$ for all n.

Lemma A.6 (Randomness passes upwards)**.** *Let α, γ be left-c.e. reals with α Martin-Löf random, let $k \ge 0$, and let $\beta := 2^{-k}\alpha + \gamma < 1$. Then $K(\alpha_{1:n}) \le K(\beta_{1:n}) + k + c$ for all n, where c depends on k and on fixed approximating programs for α and γ. Hence β is Martin-Löf random.*

Proof. Let $\alpha_s \uparrow \alpha$ and $\gamma_s \uparrow \gamma$ be computable rational approximations and $\beta_s := 2^{-k}\alpha_s + \gamma_s$. Given the string $u := \beta_{1:n}$ (which also gives n), put $q := 0.u - 2^{-n} < \beta$ and find the first s with $\beta_s > q$. Then $\beta - \beta_s < \beta - q \le 2^{1-n}$, hence $\alpha - \alpha_s \le 2^k(\beta - \beta_s) < 2^{k+1-n}$, because $\gamma - \gamma_s \ge 0$. Since α is random, it is irrational, so $\alpha_{1:n}$ is the index of the dyadic interval of length 2^{-n} that contains α, one of the at most $2^{k+1} + 1$ such intervals that meet $[\alpha_s, \alpha_s + 2^{k+1-n})$. So $\alpha_{1:n}$ is determined by u and a $(k+2)$-bit index, and $K(\alpha_{1:n}) \le K(\beta_{1:n}) + k + c$. By the Levin–Schnorr characterisation $K(\alpha_{1:n}) \ge n - c'$ for all n; hence $K(\beta_{1:n}) \ge n - k - c - c'$ for all n, and β is random by the same characterisation. □

Proof of **??**. *The machine* G_α. Let $\alpha_s \uparrow \alpha$ be a computable nondecreasing sequence of rationals, and let $G_\alpha := \{(w, \bot) : \exists s\ 0.w + 2^{-|w|} \le \alpha_s\}$, where $0.w$ is the dyadic rational with binary digits w. Then $A_\omega^{G_\alpha} \subseteq \{\bot\}$, and $\bot \in A_\omega^{G_\alpha}$ iff some prefix w of ω has $[w] \subseteq [0, \alpha_s]$ for some s, read as reals. This holds for every ω with $0.\omega < \alpha$ (take s with $\alpha_s > 0.\omega$ and a long prefix) and fails for every ω with $0.\omega > \alpha$. So the inconsistency probability of G_α is α.

The decomposition. Let p_α be a V-program for an index of G_α. Splitting the tapes by their first block in $\operatorname{dom} V$ (tapes with no such block have $A_\omega = \emptyset$, which is consistent),

$$\mathrm{Inc}_0 = 2^{-|p_\alpha|}\alpha + \gamma_\alpha, \qquad \gamma_\alpha := \sum_{p \in \operatorname{dom} V \setminus \{p_\alpha\}} 2^{-|p|}\, \mathbb{U}(O_\bot^{G_{V(p)}}),$$

and γ_α is left-c.e., because $\operatorname{dom} V \setminus \{p_\alpha\}$ is c.e. and each term is left-c.e. uniformly (Lemma A.1(a)).

(a) Let $\mathcal{H}$ be the halting set and $\alpha := \sum_{n \in \mathcal{H}} 2^{-n-1}$, which is left-c.e. Its binary expansion is the characteristic sequence of $\mathcal{H}$ (unique, since $\mathcal{H}$ is infinite and co-infinite), so α is not computable. If Inc_0 were computable, $\alpha = 2^{|p_\alpha|}(\mathrm{Inc}_0 - \gamma_\alpha)$ would be right-c.e.; being also left-c.e., it would be computable. Since a real that is both left-c.e. and right-c.e. is computable, the right-c.e. real $1 - \mathrm{Inc}_0$ is not left-c.e.; as $M_A(\emptyset) = 1 - \mathrm{Inc}_0$ is a value of M_A, M_A is not lower semicomputable.

(b) *Solovay reducibility* (Solovay, 1975; we use the definition of Downey and Hirschfeldt, 2010): $\alpha \le_S \beta$ iff there are a constant k and a partial computable function f on the rationals such that for every rational $q < \beta$, $f(q)$ is defined, $f(q) < \alpha$, and $\alpha - f(q) \le 2^k(\beta - q)$. Let $\beta := \mathrm{Inc}_0$, with approximations $\beta_s := 2^{-|p_\alpha|}\alpha_s + \gamma_{\alpha,s} \uparrow \beta$. If α is computable the reduction is trivial: given $q < \beta$, find s with $\beta_s > q$ and output a rational in $(\alpha - (\beta_s - q), \alpha)$. Otherwise $\alpha_s < \alpha$ for all s; let $f(q) := \alpha_s$ for the first s with $\beta_s > q$. Then, since $\gamma_\alpha - \gamma_{\alpha,s} \ge 0$,

$$\alpha - \alpha_s = 2^{|p_\alpha|}\big[(\beta - \beta_s) - (\gamma_\alpha - \gamma_{\alpha,s})\big] \le 2^{|p_\alpha|}(\beta - \beta_s) < 2^{|p_\alpha|}(\beta - q).$$

(c) Ω_V is Martin-Löf random (Chaitin, 1975). Apply the decomposition with $\alpha := \Omega_V$, and Lemma A.6 with $k = |p_\alpha|$ and $\gamma = \gamma_\alpha$; here $\mathrm{Inc}_0 < 1$ by **??**(e), and k and γ_α are fixed by V, so the constant of the lemma is a machine constant. □

Probabilities of other machine behaviours are studied by Becher et al. (2006) and Barmpalias et al. (2016). By Kučera and Slaman (2001) and Calude et al. (2001), Inc_0 is then an Ω-number; we do not use this.

Proposition A.7 (Computability relative to the halting problem)**.** *(a) $M_A(D)$ and the conditionals $M_A(b \mid D, \varphi)$, $M_A^3(y \mid D, \varphi)$ and $\mathrm{Bel}_D(\chi)$ are computable relative to $\emptyset'$, uniformly. (b) $D \mapsto M_A(D) + \mathrm{Inc}_0$ is lower semicomputable; so M_A is lower semicomputable relative to the single real* Inc_0.

Proof. (a) Every left-c.e. real is $\emptyset'$-computable (decide "$\exists s\ a_s > q$" for rationals q), and differences and ratios of $\emptyset'$-computable reals are $\emptyset'$-computable once the denominator is known to be positive. By **??**(a), $M_A(D) > 0$ iff D is consistent, a Π_1^0 question, which $\emptyset'$ decides. The three-valued events are Boolean combinations of effectively open classes, with $O_a \cap O_b = O_{a \wedge b}$, so their measures are sums and differences of left-c.e. reals by inclusion–exclusion. (b) is Proposition A.5. □

A.4 Dominance and invariance

Proof of **??**. $\operatorname{dom} V$ is prefix-free, so the cylinders $[p]$, $p \in \operatorname{dom} V$, are disjoint, and for $\omega = p\omega'$ we have $A_\omega = A_{\omega'}^{G_{V(p)}}$ with ω' uniform given $[p]$. Hence $\mathbb{U}(E_D \cap [p]) = 2^{-|p|}\nu_{G_{V(p)}}(D)$. The tapes

with no prefix in $\operatorname{dom} V$ have measure $1-\Omega_V$ and $A_\omega=\emptyset$, which proves ψ_D iff B does. Grouping the p with $V(p)=i$ gives the weight $\mathbf{m}(i)\geq 2^{-K(i)}$, and taking i with $\nu_{G_i}=\nu$ and $K(i)=K(\nu)$ gives $M_A\geq 2^{-K(\nu)}\nu$. The three-valued case is identical.

(i) The deterministic machine $G'_e:=\{(\epsilon,\chi):\chi\in\mathsf{W}_e\}$ has an index $g(e)$ with g computable and injective, so $\mathbf{m}(g(e))\geq 2^{-c}\mathbf{m}(e)$: the image of $\mathbf{m}$ under g is an lsc semimeasure, dominated by $\mathbf{m}$. For $e\in[T]$, the tapes beginning with a program for $g(e)$ have $A_\omega=\mathsf{W}_e$, with $\mathrm{Cn}(A_\omega)=\mathrm{Cn}(T)$, and $B\cup A_\omega$ consistent because $B\cup T$ is. These sets of tapes are disjoint for distinct e; summing gives $\mathbb{U}_A[T]\geq 2^{-c}\mathbf{m}_{\equiv}(T)$, and $\mathbf{m}_{\equiv}(T)\geq 2^{-K_{\equiv}(T)}$. The same argument over $e\in\mathrm{Ext}T$ gives $\mathbb{U}_A(\mathrm{Ext}T)\geq 2^{-c}\mathbf{m}(\mathrm{Ext}T)$.

(ii) $G_f:=\{(\epsilon,\varphi^b):f(\varphi)=b\}$ enumerates Γ_f, which is consistent with B and proves ψ_D when $f\supseteq D$; and $K(G_f)\leq K(f)+c$.

(iii) The map $i\mapsto\sum_{g(e)=i}v(e)$ is an lsc semimeasure of complexity at most $K(v)+c$, so $\mathbf{m}(g(e))\geq 2^{-K(v)-c}v(e)$. With $v=\mathbf{m}$ this gives $M_A\geq 2^{-c}W_A$.

(iv) The machine of Example (iii) in Appendix A.1 has an index computable from indices of T and of the family. □

Proposition A.8 (Choice of the universal machine)**.** *Call a theory machine G'* universal *if for every theory machine G there is a string p_G with $A^{G'}_{p_G\omega}=A^G_\omega$ for all ω. If G' is universal, then $2^{-c}M_A\leq\nu_{G'}\leq 2^cM_A$, and the same for the three-valued laws, with c depending only on G_U and G'.*

Proof. G_U is universal, with p_G a V-program for an index of G. $M_A\geq 2^{-K(G')}\nu_{G'}$ by **??**. Conversely, the tapes $p_{G_U}\omega'$ give G' the axiom set $A_{\omega'}$, so $\nu_{G'}(D)\geq 2^{-|p_{G_U}|}M_A(D)$. The three-valued laws are handled in the same way. So posteriors change by at most a constant factor, and losses in the semimeasure convention by an additive constant. □

Proposition A.9 (The background is a prior shift)**.** *Let B,B' be decidable and consistent with $B\subseteq\mathrm{Cn}_\emptyset(B')$ (plain logical consequence), and write M^B_A, $M^{B'}_A$ for SAI with these backgrounds. Then $M^B_A(D)\geq 2^{-K(B')-c}M^{B'}_A(D)$ for every D, with $K(B')$ the complexity of an index of B'; likewise $W^B_A\geq 2^{-K(B')-c}W^{B'}_A$ and $K^B_A\leq K^{B'}_A+K(B')+c$. The converse fails: a D consistent with B and inconsistent with B' has $M^{B'}_A(D)=0<M^B_A(D)$.*

Proof. The machine $G_{B'}:=\{(\epsilon,\chi):\chi\in B'\}\cup G_U$ emits $B'\cup A_\omega$ on ω. Under the background B, the set $B\cup B'\cup A_\omega$ is equivalent to $B'\cup A_\omega$, so the law of $G_{B'}$ under the background B is $M^{B'}_A$; apply **??**. (Both laws use the same G_U, which does not depend on the background.) The same map on single programs gives the statements for W_A and K_A. The converse fails as stated, since D is consistent with B (**??**(a)). □

Proof of **??**. *A preliminary fact:* $B'\vdash\neg\tau(\bot)$. By (T1) with $\Gamma=\emptyset$, $B\vdash\neg\bot$ gives $B'\vdash\tau(\neg\bot)$, and (T2) converts this.

Push forward. The machine "run G_U and emit $\tau(\chi)$ for every emitted χ" emits $\tau[A_\omega]$. By (T1), $B'\cup\tau[A]\vdash\tau(\bot)$ iff $B\cup A\vdash\bot$; with the preliminary fact, $B'\cup\tau[A]$ is consistent iff $B\cup A$ is. By (T1) and (T2), $B'\cup\tau[A]\vdash\tau(\varphi)^b$ iff $B\cup A\vdash\varphi^b$ (for $b=0$: $B'\cup\tau[A]\vdash\neg\tau(\varphi)$ iff it proves $\tau(\neg\varphi)$ iff $B\cup A\vdash\neg\varphi$). So the event of this machine for τD is exactly E_D, and **??** in $\mathcal{L}'$ gives $M'_A(\tau D)\geq 2^{-K(\tau)-c}M_A(D)$.

Pull back. For $A'\subseteq\mathrm{Sent}_{\mathcal{L}'}$ let $\tau^*(A'):=\{\varphi\in\mathrm{Sent}_{\mathcal{L}}:B'\cup A'\vdash\tau(\varphi)\}$. The set $G^*:=\{(w,\varphi):B'\cup A'_w\vdash\tau(\varphi)\}$ is c.e., and by compactness, as in Lemma A.1, $A^{G^*}_\omega=\tau^*(A'_\omega)$. If $B'\cup A'$ is consistent, so is $B\cup\tau^*(A')$: if $B\cup\Gamma\vdash\bot$ for a finite $\Gamma\subseteq\tau^*(A')$, then $B'\cup\tau[\Gamma]\vdash\tau(\bot)$ by (T1), and $B'\cup A'$ proves every member of $\tau[\Gamma]$, so $B'\cup A'\vdash\tau(\bot)\wedge\neg\tau(\bot)$. If $B'\cup A'\vdash\tau(\varphi)^b$,

then $\varphi^b \in \tau^*(A')$ (for $b = 0$ via (T2)). So the event of G^* for D contains the event for τD, and $M_A(D) \geq 2^{-K(\tau)-c} M'_A(\tau D)$.

The same two maps act on single enumerator programs, with a prefix of length $K(\tau) + c$; this gives the statements for W_A and K_A. □

Proof of **??**. (T1) holds for the inclusion because every $\mathcal{L}$-structure expands to an $\mathcal{L}'$-structure, so by completeness $B \cup \Gamma \vdash_{\mathcal{L}'} \varphi$ iff $B \cup \Gamma \vdash_{\mathcal{L}} \varphi$ for $\mathcal{L}$-sentences; (T2) is trivial. **??** with $K(\tau) = O(1)$ gives the weights, and a predictive probability is a ratio of two weights, each changed by a factor $2^{\pm c}$. □

A.5 Loss bounds

Proposition A.10 (Laws plus exceptions)**.** *(a) $L_A(D_n) \leq -\log \nu(D_n) + K(\nu)$ for every theory-machine law ν. (b) For every r.e. T and every $E \subseteq D_n$ with $B \cup T \cup E^{\pm}$ consistent and $B \cup T \cup E^{\pm} \vdash \psi_{D_n}$: $L_A(D_n) \leq K_{\equiv}(T) + K(E) + c$. (c) Let $(\alpha_j)_{j \geq 1}$ be a computable family of sentences and T r.e. with $B \cup T \cup \{\alpha_j^{x_j} : j \geq 1\}$ consistent for every $x \in \{0,1\}^{\mathbb{N}}$. If every datum of D_n is proved by T or is one of the α_j with some label, and m distinct j occur, then $L_A(D_n) \leq K(T, (\alpha_j)) + m + c$.*

?? extends (c) from atoms a law is silent about to answers it only constrains.

Proof. (a) is **??**. (b) For $e \in [T]$, the set $\mathsf{W}_e \cup \{\psi_E\}$ has an index computable from (e, E); it is consistent with B, since $B \cup T \cup E^{\pm}$ is, and it proves ψ_{D_n}. As $K(e, E) \leq K(e) + K(E) + c$, **??**(i) gives $L_A(D_n) \leq K(e) + K(E) + c'$; minimise over $e \in [T]$. (c) Consider the machine that emits T and $\alpha_j^{\omega_j}$ for every j. By the hypothesis its theory is consistent on every tape, and T decides no α_j. Its theory proves every datum proved by T, and proves the α_j-data whenever ω agrees with their m labels, which has probability 2^{-m}. The machine has an index computable from those of T and of the family; apply (a). □

Proposition A.11 (Mistake bounds)**.** *(a) Under the hypotheses of* **??***, the predictor that outputs the most probable of $1, 0, \iota$ under $(\mathrm{Bel}, \mathrm{Dis}, \mathrm{Indep})_{D_{t-1}}(\varphi_t)$ (ties broken arbitrarily; the stored label at a repeat) makes at most $-\log(\mathbb{U}_A(\mathrm{Ext}T)/M_A(\emptyset)) \leq K_{\equiv}(T) + c$ mistakes, a mistake being any output other than b_t, ι included. (b) Fix a well-order of $\mathrm{dom}\, V$ that refines length. The* Kolmogorov predictor *takes at step t the first program p_t with $B \cup \mathsf{W}_{V(p_t)}$ consistent and proving $\psi_{D_{t-1}}$, predicts 1 (resp. 0) if $\mathsf{W}_{V(p_t)}$ proves φ_t (resp. $\neg\varphi_t$), and abstains otherwise. If $k := \sup_n K_A(D_n) < \infty$ (for instance $k \leq K_{\equiv}(T) + c$ as in* **??***), it errs or abstains at most 2^k times. (c) The order hypothesis in (b) is needed: for every m there are a well-order of $\mathrm{dom}\, V$ of type ω and a data set D with $K_A(D) \leq K(m) + c \leq 2 \log m + c'$ on which the Kolmogorov predictor with that order errs or abstains $2^m - 1$ times.*

The bound of (b) is essentially attained even for length-refining orders, on data where the Bayes predictor makes at most $K_A(D) + c$ mistakes (Theorem A.35): it is the selection rule that fails, not the prior.

Proof of (a) and (b). (a) At a repeat the predictor outputs the stored label and makes no mistake. Otherwise the posterior trichotomy sums to 1, so if $M_A(b_t \mid D_{t-1}, \varphi_t) = \mathrm{Bel}_{D_{t-1}}(\varphi_t^{b_t}) > \frac{1}{2}$, then b_t is the unique maximiser. Each mistake therefore costs at least one bit of cumulative loss in the semimeasure convention, and by **??** the cumulative loss is $L_A(D_n) - L_A(\emptyset) \leq -\log \mathbb{U}_A(\mathrm{Ext}T) + \log M_A(\emptyset)$. This is the halving argument of Corollary 5.4 of the companion paper (Hänni and Claude (Anthropic), 2026b).

(b) At an error or abstention, p_t does not prove $\varphi_t^{b_t}$, so it is incompatible with D_t and with every later $D_{t'}$, and it is never chosen again. (At a repeat p_t proves the stored label, since it is compatible with D_{t-1}.) Since the order refines length, each p_t is a shortest compatible program, of length $K_A(D_{t-1}) \le k$. Kraft's inequality for the prefix-free set $\operatorname{dom} V$ allows at most 2^k programs of length at most k. □

Part (c) is proved in Appendix A.9, after Theorem A.35.

Theorem A.12 (Merging for three-valued data)**.** *Let G be a theory machine with $\nu_G(\emptyset) = 1$, $\omega \sim \mathbb{U}$ and $A := A_\omega^G$. Let (u_t) be random variables independent of ω, with any joint law, and let the questions be $\varphi_t := s_t(h_{t-1}, u_t)$ for arbitrary measurable rules s_t, where $h_{t-1} = (\varphi_1, y_1, \dots, \varphi_{t-1}, y_{t-1})$ and $y_t \in Y_3$ is the trichotomy label of φ_t under A. Let $D_t := \{(\varphi_s, y_s) : s \le t\}$ and $d_t := \mathrm{KL}(\nu_G^3(\cdot \mid D_{t-1}, \varphi_t) \,\|\, M_A^3(\cdot \mid D_{t-1}, \varphi_t))$ in bits (0 at a repeat). Then for every n, $\sum_{t \le n} \mathbb{E} d_t = \mathbb{E} \log \frac{\nu_G^3(D_n)}{M_A^3(D_n)} + \log M_A(\emptyset) \le K(G) + \log M_A(\emptyset) < K(G)$. Hence $d_t \to 0$ almost surely, and $\sum_t \mathbb{E} \|\nu_G^3(\cdot \mid D_{t-1}, \varphi_t) - M_A^3(\cdot \mid D_{t-1}, \varphi_t)\|_1^2 \le 2 \ln 2 \cdot K(G)$.*

With two-valued data the question rule would have to see A for the two-valued conditionals to be the data law, and the expected bound can fail (as in the remark after Theorem 5.3 of the companion paper); the pathwise bound of Proposition A.10(a) still holds.

Proof. Step 1: the conditional law of the next label. Histories take countably many values. For a value h of the history and a value of $u_{<t}$, the event $\{h_{t-1} = h\}$ is the intersection of an event in $u_{<t}$ (the rules produce the questions of h from its labels) with the event $\{\mathrm{tri}_\omega^G \supseteq D_h\}$ (**??**(g)), where D_h is the data set of h. Since $\nu_G(\emptyset) = 1$, the theory is consistent and tri_ω^G is defined on almost every tape. Since (u_t) is independent of ω, given $(h_{t-1}, u_{\le t})$ the tape has the law $\mathbb{U}(\cdot \mid \mathrm{tri}_\omega^G \supseteq D_{t-1})$, and $y_t = \mathrm{tri}_\omega^G(\varphi_t)$ has the law $\nu_G^3(\cdot \mid D_{t-1}, \varphi_t)$. (Almost surely $\nu_G^3(D_{t-1}) > 0$.)

Step 2: telescoping. Let $Z_t := \log \nu_G^3(y_t \mid D_{t-1}, \varphi_t) - \log M_A^3(y_t \mid D_{t-1}, \varphi_t)$ (0 at a repeat) and $S_n := \sum_{t \le n} Z_t$. Since $\nu_G^3(\emptyset) = 1$ and $M_A^3(\emptyset) = M_A(\emptyset)$, $S_n = \log \nu_G^3(D_n) - \log M_A^3(D_n) + \log M_A(\emptyset)$.

Step 3: integrability. By **??**, $M_A^3 \ge 2^{-K(G)} \nu_G^3$, so $S_n \le K(G) + \log M_A(\emptyset)$ pointwise. Since $-\log M_A^3(D_n) \ge 0$, $S_n \ge \log \nu_G^3(D_n) + \log M_A(\emptyset)$. And $\mathbb{E}[-\log \nu_G^3(D_n)] = \sum_{t \le n} \mathbb{E}[H_t] \le n \log 3$, where H_t is the entropy of $\nu_G^3(\cdot \mid D_{t-1}, \varphi_t)$, a distribution on three points: the summands $-\log \nu_G^3(y_t \mid D_{t-1}, \varphi_t)$ are nonnegative, so the tower property applies by Step 1 (Tonelli). Hence S_n is integrable, and so is each $Z_t = S_t - S_{t-1}$.

Step 4. By Step 1, $\mathbb{E}[Z_t \mid h_{t-1}, u_{\le t}] = d_t$, which is nonnegative because the conditionals of M_A^3 are probability distributions (**??**(f)). Taking expectations and summing gives $\sum_{t \le n} \mathbb{E} d_t = \mathbb{E} S_n$, which is the identity and the bound. Pinsker's inequality in bits, $\mathrm{KL} \ge \|P - Q\|_1^2 / (2 \ln 2)$, gives the last claim, and $\mathbb{E} \sum_t d_t < \infty$ gives $d_t \to 0$ almost surely. □

Corollary A.13 (Three-valued loss)**.** *If D_n is three-valued and labelled by the trichotomy of an r.e. T with $B \cup T$ consistent, then in any reveal order the loss in the three-valued convention is $-\log M_A^3(D_n) + \log M_A(\emptyset) \le -\log \mathbb{U}_A[T] \le K_{\equiv}(T) + c$.*

Proof. The tapes with $B \cup A_\omega$ consistent and $\mathrm{Cn}(A_\omega) = \mathrm{Cn}(T)$ lie in $E_{D_n}^3$. Order independence holds for M_A^3 with the proof of **??**(d), using (f) in place of (b); then apply **??**(i). Here $[T]$ replaces $\mathrm{Ext}T$: three-valued data penalise theories that decide more than T. □

A.6 Coding bounds and the absence of a coding theorem

Lemma A.14 (Literals of a fresh predicate)**.** *In the literal setting: (L1) every set Λ of literals $R(t_w)^b$ with at most one literal per word is consistent with B; (L2) if $B \cup \Lambda$ is consistent, then $B \cup \Lambda \vdash R(t_y)^b$ only if $R(t_y)^b \in \Lambda$. The same holds for finitely many fresh predicates. Moreover a propositional combination of fresh atoms is consistent with B iff it is propositionally satisfiable, and B together with such combinations proves another such combination iff they imply it propositionally.*

Proof. (L1) Expand the model of B in which the t_w are pairwise distinct by interpreting R as the set of the elements t_w with $R(t_w) \in \Lambda$; B does not mention R, so it stays true. (L2) If $R(t_y)^{1-b} \in \Lambda$, a proof of $R(t_y)^b$ would make $B \cup \Lambda$ inconsistent. If neither literal of t_y is in Λ, then $\Lambda \cup \{R(t_y)^{1-b}\}$ has at most one literal per word and is consistent by (L1). For the last sentence: every valuation of finitely many fresh atoms is realised in an expansion of the model of B, so a propositional combination of fresh atoms is consistent with B iff it is propositionally satisfiable, and the implication statement follows by applying this to the premises together with the negated conclusion. □

Proof of **??**. (a) Let $Q_S(\sigma)$ be the probability, over ω, that the following procedure outputs σ: at stage s, read $\omega_{1:s}$, run s steps of the enumeration of G_U on it, and test all derivations of size at most s from B and the axioms emitted so far, for every member of S in a fixed order; output the first member found. The output events are disjoint open sets, so $\sum_{\sigma \in S} Q_S(\sigma) \leq 1$, and $Q_S(\sigma)$ is lsc uniformly in (S, σ). If $B \cup A_\omega$ is consistent and proves σ, it proves no other member of S, so the procedure outputs σ; hence $\mathrm{Bel}_0(\sigma) \leq Q_S(\sigma)$. The conditional coding theorem (Section 1.1) gives $K(\sigma \mid S) \leq -\log Q_S(\sigma) + c$. (So the coding theorem is applied to Q_S, which is lsc, and not to M_A, which is not; **??**.)

(b) Apply (a) with S the label conjunctions on $\mathrm{dom}\, D$, which are pairwise logically incompatible and computable from $\mathrm{dom}\, D$, and $\sigma := \psi_D$, so that $M_A(D) = \mathrm{Bel}_0(\psi_D)$ (**??**(c)). From $\mathrm{dom}\, D$ and ψ_D one computes b_D, so $K(b_D \mid \mathrm{dom}\, D) \leq K(\psi_D \mid S) + c$.

(c) The first inequality is one term of W_A; the second is **??**(iii). For the third, $K_A(D) \leq K(\psi_D) + c$ (Remark A.2(c)) $\leq K(S) + K(\psi_D \mid S) + c'$, and (a) bounds $K(\psi_D \mid S)$ by $L_A(D) + c$. □

Proposition A.15 (Lower bound)**.** *$L_A(D) \geq K(b_D \mid \mathrm{dom}\, D) - c$ for every D, where b_D is the vector of two-valued labels in canonical order. So in the semimeasure convention, on fair-coin labels of any fixed n sentences, $\mathbb{E} L_A \geq n - c$.*

Proof. The inequality is **??**(b). For the expectation, the weights $2^{-K(b \mid \mathrm{dom}\, D)}$ sum to at most 1 (Kraft's inequality), so Gibbs' inequality gives $\mathbb{E}_b K(b \mid \mathrm{dom}\, D) \geq n$ for uniform $b \in \{0,1\}^n$. With logically dependent sentences, inconsistent labellings have $L_A = \infty$, and the bound holds trivially for them. □

Proof of **??**. (i) Let $\ell := \ell_d$, so that $2^\ell < d/\log 3$ and $3^{2^\ell} < 2^d$. There are fewer than 2^ℓ programs of length less than ℓ. Colour each $x \in \{0,1\}^d$ by the vector, over these programs p, of the following symbols: "+" if $B \cup \mathsf{W}_{V(p)}$ is consistent and proves $R(t_x)$; "−" if it is consistent and proves $\neg R(t_x)$; "∗" otherwise. There are at most $3^{2^\ell - 1} < 2^d$ colours, so two distinct words x_1, x_2 share a colour. A program p with $|p| < \ell$ that is compatible with D would colour x_1 "−" and x_2 "+". So every compatible program has length at least ℓ, which gives both claims of (i); and $\ell \geq \log(d/\log 3) - 1 > \log d - 2$, since $\log\log 3 < 0.67$.

(ii) Let $j(x)$ be the position of x in the length-lexicographic order ($j(\epsilon) = 0$), so that $j(x) \leq 2^{d+1} - 2$ for $|x| \leq d$. Let $s := \lceil \log(d+1) \rceil$ and $L := 2^{2^s} \geq 2^{d+1}$. For $u \in \{0,1\}^L$ let $\Lambda_u :=$

$\{R(t_x)^{u_{j(x)+1}} : j(x) < L\}$; it is consistent by (L1) of Lemma A.14 and proves $R(t_x)^b$ iff $u_{j(x)+1} = b$, by (L2). A fixed prefix machine reads a V-program for s, computes L, reads L further bits u and outputs an index of Λ_u; by universality of V a fixed string r makes $r\,q\,u \in \operatorname{dom} V$ a program for that index, for q a shortest program for s. These 2^L programs are distinct, form a prefix-free family (since q fixes L), and have length $|r| + K(s) + L$. For a two-point D on distinct words of length at most d, exactly a quarter of the strings u give a compatible Λ_u. So $W_A(D) \geq 2^L \cdot \frac{1}{4} \cdot 2^{-|r|-K(s)-L}$.

(iii) The theory machine $G_{\mathrm{coin}} := \{(w, R(t_x)^{w_{j(x)+1}}) : |w| = j(x)+1\}$ emits $\{R(t_x)^{\omega_{j(x)+1}} : x \in \{0,1\}^*\}$, which is consistent by (L1); its theory proves D iff ω agrees with the labels of D at two positions (L2), which has probability $\frac{1}{4}$. Apply **??**.

The consequences: $K_A + \log W_A \geq \ell_d - 2 - K(s) - c$, and $\Omega_V^{\geq \ell}$ is the tail of a convergent series, with $\ell_d \to \infty$. □

The bounds of **??** are stated with $K_{\equiv}(T)$, the least complexity of an index of a theory equivalent to T (Definition 1.8); for whole theories the question is how $K_{\equiv}$ compares with the mixture weight $\mathbf{m}_{\equiv}$. For an r.e. set $Y \subseteq \mathrm{Sent}$ let $K_{\mathrm{enum}}(Y) := \min\{|p| : p \in \operatorname{dom} V,\ \mathsf{W}_{V(p)} = Y\}$ and $\mathbf{m}_{\mathrm{enum}}(Y) := \sum\{2^{-|p|} : p \in \operatorname{dom} V,\ \mathsf{W}_{V(p)} = Y\}$, its enumeration complexity and the corresponding discrete semimeasure. A theorem of Solovay (Solovay, 1977), in the form stated by Vereshchagin (2007), says that $K_{\mathrm{enum}}(Y) \leq 3I(Y) + O(\log I(Y))$ for every r.e. set Y, where $I(Y)$ is the negative logarithm of the probability that a machine with a random input tape enumerates exactly Y.

Proposition A.16 (Coding for whole theories)**.** *(a) For every r.e. T: $K_{\equiv}(T) = K_{\mathrm{enum}}(\mathrm{Cn}T) \pm c$ and $2^{-c}\mathbf{m}_{\mathrm{enum}}(\mathrm{Cn}T) \leq \mathbf{m}_{\equiv}(T) \leq 2^c \mathbf{m}_{\mathrm{enum}}(\mathrm{Cn}T)$. (b) For every r.e. T: $-\log \mathbf{m}_{\equiv}(T) \leq K_{\equiv}(T) \leq 3\,(-\log \mathbf{m}_{\equiv}(T)) + O\big(\log(-\log \mathbf{m}_{\equiv}(T))\big) + c$.*

Proof. (a) Let κ be computable and injective with $\mathsf{W}_{\kappa(e)} = \mathrm{Cn}(\mathsf{W}_e)$ (the s-m-n theorem with padding). Every e' with $\mathsf{W}_{e'} = \mathrm{Cn}T$ lies in $[T]$, so $K_{\equiv}(T) \leq K_{\mathrm{enum}}(\mathrm{Cn}T) + c$ and $\mathbf{m}_{\equiv}(T) \geq 2^{-c}\mathbf{m}_{\mathrm{enum}}(\mathrm{Cn}T)$ (grouping programs by the index they print). Conversely κ maps $[T]$ into $\{e' : \mathsf{W}_{e'} = \mathrm{Cn}T\}$, with $K(\kappa(e)) \leq K(e) + c$ and $\mathbf{m}(\kappa(e)) \geq 2^{-c}\mathbf{m}(e)$, since the image of $\mathbf{m}$ under κ is an lsc semimeasure. (b) The left inequality holds because $2^{-K_{\equiv}(T)}$ is one term of $\mathbf{m}_{\equiv}(T)$. For the right one, the probability that a machine with a random input tape enumerates exactly Y is at least $2^{-c}\mathbf{m}_{\mathrm{enum}}(Y)$, since a deterministic program is a randomised algorithm that ignores its tape; so Solovay's inequality also holds with $-\log \mathbf{m}_{\mathrm{enum}}(Y)$ in place of $I(Y)$. Apply it to $Y = \mathrm{Cn}T$ and use (a). □

Remark A.17 (Factor two). Vereshchagin (2007) replaces Solovay's constant 3 by 2 for finite sets, and Shekhovtsov and Zakharov (2023) show that a bound of this kind for finite sets transfers to all sets up to logarithmic terms. So the factor 3 in Proposition A.16(b) and in Proposition A.18(c) can be replaced by 2, up to logarithmic terms whose form we do not state.

Whether the random tape helps for whole theories is the question whether a probabilistic enumerator of a set can be much more likely than any short deterministic one. Let $\mathbb{U}_{\mathrm{enum}}(Y) := \mathbb{U}\{\omega : A_\omega = Y\}$, the probability that the universal theory machine emits exactly Y; for sets of strings, let $\mathbb{U}^{\mathrm{str}}_{\mathrm{enum}}$, $\mathbf{m}^{\mathrm{str}}_{\mathrm{enum}}$ and $K^{\mathrm{str}}_{\mathrm{enum}}$ be the same quantities for a universal probabilistic enumerator of sets of strings and for programs of V printing indices of r.e. sets of strings.

Proposition A.18 (The random tape and whole theories)**.** *(a) For every r.e. T with $B \cup T$ consistent: $2^{-c}\mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T) \leq \mathbb{U}_A[T] \leq 2^c \mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T)$, and $\mathbb{U}_A[T] \geq 2^{-c}\mathbf{m}_{\equiv}(T)$. (b) In the literal setting: $\mathbb{U}_A[T] \asymp \mathbf{m}_{\equiv}(T)$ for all consistent r.e. T iff $\mathbb{U}^{\mathrm{str}}_{\mathrm{enum}}(Y) \asymp \mathbf{m}^{\mathrm{str}}_{\mathrm{enum}}(Y)$ for all r.e.*

$Y \subseteq \{0,1\}^*$*; and* $K_{\equiv} \le -\log \mathbf{m}_{\equiv} + O(1)$ *iff* $K^{\mathrm{str}}_{\mathrm{enum}} \le -\log \mathbf{m}^{\mathrm{str}}_{\mathrm{enum}} + O(1)$. *(c)* $-\log \mathbb{U}_A[T] \le -\log \mathbf{m}_{\equiv}(T) + c \le K_{\equiv}(T) + c \le 3\,(-\log \mathbb{U}_A[T]) + O\big(\log(-\log \mathbb{U}_A[T])\big) + c$ *for every r.e.* T *with* $B \cup T$ *consistent.*

Proof. (a) $\{\omega : A_\omega = \mathrm{Cn}T\} \subseteq \{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}T\}$, and both sets consist of consistent theories, so $\mathbb{U}_A[T] \ge \mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T)$. Conversely, the machine G_{cl} that runs G_U and emits $\mathrm{Cn}(A_\omega)$ emits exactly $\mathrm{Cn}T$ on $\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}T\}$, and the tapes $p_{G_{\mathrm{cl}}}\omega'$, with $p_{G_{\mathrm{cl}}}$ a V-program for an index of G_{cl}, show $\mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T) \ge 2^{-|p_{G_{\mathrm{cl}}}|}\mathbb{U}_A[T]$. The last claim is **??**(i).

(b) ($\Leftarrow$) By (a) and Proposition A.16(a), $\mathbb{U}_A[T] \asymp \mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T)$ and $\mathbf{m}_{\equiv}(T) \asymp \mathbf{m}_{\mathrm{enum}}(\mathrm{Cn}T)$, and coding sentences as strings changes the enumeration quantities by constant factors. ($\Rightarrow$) For $Y \subseteq \{0,1\}^*$ let $T_Y := \{R(t_x) : x \in Y\}$. By (L2) of Lemma A.14, $Y = \{x : R(t_x) \in \mathrm{Cn}(T_Y)\}$. A probabilistic enumerator of exactly Y yields one of exactly $\mathrm{Cn}(T_Y)$ (close under Cn) and conversely (extract the words x), with constant overhead, for probabilistic and for deterministic programs alike; and distinct Y give distinct $\mathrm{Cn}(T_Y)$. So $\mathbb{U}^{\mathrm{str}}_{\mathrm{enum}}(Y) \asymp \mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T_Y) \asymp \mathbb{U}_A[T_Y]$ and $\mathbf{m}^{\mathrm{str}}_{\mathrm{enum}}(Y) \asymp \mathbf{m}_{\equiv}(T_Y)$; likewise for the complexities.

(c) The first two inequalities are (a) and $\mathbf{m}_{\equiv}(T) \ge 2^{-K_{\equiv}(T)}$. For the last, Solovay's theorem bounds $K_{\mathrm{enum}}(Y)$ by $3I(Y) + O(\log I(Y))$, and $I(Y) = -\log \mathbb{U}_{\mathrm{enum}}(Y) \pm c$ for the universal probabilistic enumerator; apply it to $Y = \mathrm{Cn}T$, and use (a) and Proposition A.16(a). □

Question A.19. Does the random tape help for whole theories? That is, is $\mathbb{U}_A[T] \asymp \mathbf{m}_{\equiv}(T)$ for every consistent r.e. T, and is $K_{\equiv}(T) \le -\log \mathbf{m}_{\equiv}(T) + O(1)$? By Proposition A.18(b) these are, in the literal setting, the questions whether randomness helps to enumerate a set of strings.

The question matters for identification: the limit loss on a complete presentation of T^* is $-\log \mathbb{U}_A[T^*] + \log M_A(\emptyset)$ (Proposition A.27(a)), and Question A.19 asks whether this is $-\log \mathbf{m}_{\equiv}(T^*)$ up to a constant.

A.7 The consistency-requiring variants

Definition A.20 (Graded consistency-requiring weight)**.** For consistent A let $\mathrm{gr}_A(\varphi) := 1$ if $B \cup A \vdash \varphi^{D(\varphi)}$ and $\mathrm{gr}_A(\varphi) := \frac{1}{2}$ otherwise, and

$$\nu^{1/2}_A(D) := \mathbf{1}[B \cup A \cup D^{\pm} \text{ consistent}] \prod_{\varphi \in \mathrm{dom}\, D} \mathrm{gr}_A(\varphi), \qquad M^{1/2}_A(D) := \int_{\mathrm{Cons}} \nu^{1/2}_{A_\omega}(D)\, d\mathbb{U}(\omega).$$

The graded weight is the symmetric form of the graded score S_g of Definition 2.12 of the earlier report with $g \equiv 1$ on proved data and $g_\infty = \frac{1}{2}$: S_g grades only the data labelled true, $\nu^{1/2}$ grades the data labelled false too. The results below use only data labelled true, where the two agree.

Throughout, $H_\omega := B \cup A_\omega \cup D^{\pm}$; when $M^{\mathrm{nc}}_A(D) > 0$, write $\Pr_D := \mathbb{U}(\cdot \mid H_\omega \text{ consistent})$, so that $\mathrm{Bel}^{\mathrm{nc}}_D(\varphi) = \Pr_D[H_\omega \vdash \varphi]$.

Proof of **??**. (a) For consistent H, the set $H \cup \{\varphi^b\}$ is consistent iff $H \nvdash \varphi^{1-b}$; so both labels are possible iff H decides neither φ nor $\neg\varphi$, and exactly one is possible otherwise; for inconsistent H neither is. That $M^{\mathrm{nc}}_A(D) = \mathrm{Pl}_0(\psi_D)$ is Proposition A.4. (b) If $A_\omega = \emptyset$, then $H_\omega = B \cup D^{\pm}$ is consistent; and these tapes include those beginning with $p_\emptyset$. For the coin-free and Kolmogorov versions, the program $p_\emptyset$ is a witness on every consistent D. (c) holds by definition. □

Proposition A.21 (The consistency-requiring predictor)**.** *(a) For every consistent D and $\varphi \notin \operatorname{dom} D$ with $B \cup D^{\pm} \nvdash \varphi^b$, $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi^b) \le 1 - \varepsilon_\emptyset / M_A^{\mathrm{nc}}(D) \le 1 - \varepsilon_\emptyset$. So on every consistent reveal sequence the cumulative loss $\sum_t -\log \mathrm{Bel}_{D_{t-1}}^{\mathrm{nc}}(\varphi_t^{b_t})$ is at least $\log \frac{1}{1-\varepsilon_\emptyset}$ times the number of steps whose datum is not entailed by the earlier data. (b) In the literal setting, let $a_i, s_{i,j}$ $(i, j \ge 1)$ be fresh atoms, computable in (i, j), $\zeta_{i,j} := \neg s_{i,1} \wedge \cdots \wedge \neg s_{i,j-1} \wedge s_{i,j}$, $b_{i,j} := a_i \vee \zeta_{i,j}$, and $c_R := |p_R|$ for a V-program p_R of an index of the deterministic machine emitting $\forall x\, R(x)$. (b1) For $D_j := \{(a_1, 1), (b_{1,j}, 1)\}$ and all but at most $1/(\mathrm{Bel}_0(a_1)\varepsilon_\emptyset)$ indices j, the consistency-requiring probability of D_j is strictly larger in the order $(a_1, b_{1,j})$ than in the order $(b_{1,j}, a_1)$. (b2) For every n there are $j_1, \ldots, j_n \le \lceil 2^{c_R+1}/\varepsilon_\emptyset \rceil$ such that, for $D_n := \{(a_i, 1), (b_{i,j_i}, 1) : i \le n\}$, the cumulative loss in the order $(b_{1,j_1}, a_1, b_{2,j_2}, a_2, \ldots)$ exceeds that in the order $(a_1, b_{1,j_1}, a_2, b_{2,j_2}, \ldots)$ by at least $n \log(1 + 2^{-c_R}\varepsilon_\emptyset/2)$ bits.*

On a stream $(\alpha_j, 1)_{j \ge 1}$ of fresh atoms no datum is entailed, so (a) gives a loss linear in n, while $\{\forall x\, R(x)\}$ proves the stream and the proof-requiring loss is $O(1)$ (**??**). In (b) revealing the stronger datum first is more probable; under normalised proof-requiring prediction the weaker datum first is (Theorem A.33).

Proof. (a) The tapes with $A_\omega = \emptyset$, of mass $\varepsilon_\emptyset$, lie in the event $\{H_\omega \text{ consistent}\}$ that normalises $\mathrm{Bel}_D^{\mathrm{nc}}$, and not in its numerator, since $B \cup D^{\pm} \nvdash \varphi^b$.

(b) *Step 1: one pair, after any revealed set.* Let D be consistent with $M_A^{\mathrm{nc}}(D) > 0$, let a, ζ be sentences and $b := a \vee \zeta$; write $\Pr := \Pr_D$ and $H := H_\omega$. Put

$$\kappa := \Pr[H \cup \{b\} \text{ consistent}], \qquad \theta := \Pr[H \cup \{b\} \text{ consistent},\ H \vdash b \to a],$$
$$r := \Pr[H \cup \{\zeta \wedge \neg a\} \text{ consistent}], \quad q := \Pr[H \vdash b,\ H \nvdash a].$$

The consistency-requiring probability of revealing $(a, 1)$ and then $(b, 1)$ after D is $P_{(a,b)} := \mathrm{Bel}_D^{\mathrm{nc}}(a)\mathrm{Bel}_{D+(a,1)}^{\mathrm{nc}}(b) = \Pr[H \vdash a]$, because $a \vdash b$, so that $H \cup \{a\}$ proves b whenever it is consistent. The probability of the other order is $P_{(b,a)} := \mathrm{Bel}_D^{\mathrm{nc}}(b)\mathrm{Bel}_{D+(b,1)}^{\mathrm{nc}}(a) = \Pr[H \vdash b] \cdot \theta/\kappa$: by the deduction theorem $H \cup \{b\} \vdash a$ iff $H \vdash b \to a$, and the ratio of the $\mathbb{U}$-measures of two events inside $\{H \text{ consistent}\}$ equals the ratio of their Pr-measures. Two identities hold. First, $\kappa - \theta = r$: $H \nvdash b \to a$ iff $H \cup \{b, \neg a\}$ is consistent, which implies that $H \cup \{b\}$ is consistent, and $\{b, \neg a\}$ is equivalent to $\{\zeta \wedge \neg a\}$. Second, $\Pr[H \vdash b] = \Pr[H \vdash a] + q$, since $H \vdash a$ implies $H \vdash b$. If $\Pr[H \vdash a] > 0$, then $\theta \ge \Pr[H \vdash a] > 0$ and $\Pr[H \vdash b] > 0$, and substituting,

$$\frac{P_{(a,b)}}{P_{(b,a)}} = \frac{\Pr[H \vdash a]\,(\theta + r)}{(\Pr[H \vdash a] + q)\,\theta} = 1 + \frac{z}{\Pr[H \vdash b]\,\theta}, \qquad z := \Pr[H \vdash a]\, r - q\,\theta \ \ge\ \Pr[H \vdash a]\, r - q.$$

Step 2: disjointness. For fixed i and D, the events $\{H \vdash b_{i,j},\ H \nvdash a_i\}$ are pairwise disjoint in j. Indeed $\zeta_{i,j}$ and $\zeta_{i,j'}$ $(j < j')$ are incompatible, since $\zeta_{i,j}$ asserts $s_{i,j}$ and $\zeta_{i,j'}$ denies it; so $H \vdash a_i \vee \zeta_{i,j}$ and $H \vdash a_i \vee \zeta_{i,j'}$ give $H \vdash a_i \vee (\zeta_{i,j} \wedge \zeta_{i,j'})$, hence $H \vdash a_i$. So $\sum_j q_{i,j} \le 1$, where $q_{i,j}$ is q for the pair $(a_i, b_{i,j})$.

Step 3: lower bounds. Let D be $\emptyset$ (for (b1)) or $D_{i-1} := \{(a_{i'}, 1), (b_{i',j_{i'}}, 1) : i' < i\}$ (for (b2)); then $D^{\pm}$ is implied by $\{a_{i'} : i' < i\}$.

- $r_{i,j} \ge \varepsilon_\emptyset$. On the tapes with $A_\omega = \emptyset$ we have $H = B \cup D^{\pm}$, and $H \cup \{\zeta_{i,j} \wedge \neg a_i\}$ is consistent by (L1) of Lemma A.14: it is implied by the literal set $\{a_{i'} : i' < i\} \cup \{\neg a_i\} \cup \{\text{the literals of } \zeta_{i,j}\}$, which has one literal per word. These tapes have Pr-mass $\varepsilon_\emptyset / M_A^{\mathrm{nc}}(D) \ge \varepsilon_\emptyset$.
- $\Pr[H \vdash a_i] \ge 2^{-c_R}$ for (b2). On the tapes beginning with p_R, $H = B \cup \{\forall x\, R(x)\} \cup D^{\pm}$ is consistent (B does not mention R, so interpret R as the whole domain; $D^{\pm}$ consists of the $a_{i'}$ and disjunctions containing them) and proves a_i; their Pr-mass is $2^{-c_R}/M_A^{\mathrm{nc}}(D) \ge 2^{-c_R}$.

- For (b1), $D = \emptyset$, $\Pr = \mathbb{U}(\cdot \mid \mathrm{Cons})$ and $\Pr[H \vdash a_1] = \mathrm{Bel}_0(a_1)/(1 - \mathrm{Inc}_0) \geq \mathrm{Bel}_0(a_1) > 0$, by the one-axiom theory $\{a_1\}$.

Conclusion. (b1) By Steps 1 and 3, $z_j \geq \mathrm{Bel}_0(a_1)\varepsilon_\emptyset - q_{1,j}$, which is positive unless $q_{1,j} \geq \mathrm{Bel}_0(a_1)\varepsilon_\emptyset$; by Step 2 that happens for at most $1/(\mathrm{Bel}_0(a_1)\varepsilon_\emptyset)$ indices j. For the others $P_{(a,b)}/P_{(b,a)} > 1$. (b2) Choose $j_1, j_2, \ldots$ in turn. At step i, with $D = D_{i-1}$ and $J := \lceil 2^{c_R+1}/\varepsilon_\emptyset \rceil$, Step 2 gives some $j_i \leq J$ with $q_{i,j_i} \leq 1/J \leq 2^{-c_R}\varepsilon_\emptyset/2$, and then $z_i \geq 2^{-c_R}\varepsilon_\emptyset - 2^{-c_R}\varepsilon_\emptyset/2 = 2^{-c_R}\varepsilon_\emptyset/2$ by Step 3. Since $0 < \Pr[H \vdash b]\,\theta \leq 1$, $\log(P_{(a,b)}/P_{(b,a)}) \geq \log(1 + z_i) \geq \log(1 + 2^{-c_R}\varepsilon_\emptyset/2)$. The conditionals of the consistency-requiring predictor depend on the revealed data only as a set, and after $i - 1$ completed pairs both orders have revealed D_{i-1}. So the loss of the order with b first minus that of the order with a first is $\sum_{i \leq n} \log(P^{(i)}_{(a,b)}/P^{(i)}_{(b,a)}) \geq n \log(1 + 2^{-c_R}\varepsilon_\emptyset/2)$. □

For a single pair the sign of z is not determined by universality; the counting over j in Step 2 is what makes (b) hold for M_A itself. The streams of (b2) are not computable (the j_i depend on M_A); they use at most J candidate atoms per block. The consistency-requiring posterior is order independent, since it depends on D as a set; its predictive probabilities do not chain, because $\mathrm{Bel}^{\mathrm{nc}}_D(\varphi) \neq M^{\mathrm{nc}}_A(D + (\varphi, 1))/M^{\mathrm{nc}}_A(D)$ in general.

An explicit law. For a single law the order effect is easy to see. Let a, χ be fresh atoms and $G := \{(1, a)\}$, whose theory is $\{a\}$ on the tapes beginning with 1 and $\emptyset$ otherwise, and let $D := \{(a, 1), (a \vee \chi, 1)\}$. In the order $(a, a \vee \chi)$ the consistency-requiring probability of D under G is $\frac{1}{2}$: only $\{a\}$ proves a, and then $a \vee \chi$ is entailed. In the order $(a \vee \chi, a)$ it is $\frac{1}{4}$: the first factor is $\frac{1}{2}$, and given $a \vee \chi$ both theories remain consistent and only $\{a\} \cup \{a \vee \chi\}$ proves a, so the second factor is $\frac{1}{2}$.

Proposition A.22 (Graded consistency-requiring weight)**.** *(a) For consistent A, $\nu_A^{1/2}$ is an order-independent weak semimeasure with $\nu_A^{1/2}(\emptyset) = 1$; hence $M_A^{1/2}$ is a weak semimeasure with $M_A^{1/2}(\emptyset) = 1 - \mathrm{Inc}_0$, and the conditionals of $M_A^{1/2}/M_A^{1/2}(\emptyset)$ form an order-independent predictor. (b) $M_A^{1/2} \geq M_A$. (c) For fresh atoms p, q and $D := \{(p, 1)\}$, $M_A^{1/2}(D + (p \vee q, 1)) \leq M_A^{1/2}(D) - \varepsilon_\emptyset/4$, although $B \cup D^{\pm} \vdash p \vee q$; so $M_A^{1/2}$ is not the law of any theory machine. (d) $M_A(\{(p \wedge q, 1)\}) = M_A(\{(p, 1), (q, 1)\})$, but $M_A^{1/2}(\{(p \wedge q, 1)\}) - M_A^{1/2}(\{(p, 1), (q, 1)\}) \geq \varepsilon_\emptyset/4$. (e) $M_A^{1/2}/M_A^{1/2}(\emptyset)$ is not a containment functional, not even of a random partial function on* Sent *that ignores logic: for every φ, its Möbius sum on the inputs $\varphi, \neg\varphi$ (Proposition A.2 of the companion paper) equals $-\frac{1}{2}\mathbb{U}\{\omega \in \mathrm{Cons} : A_\omega$ decides neither φ nor $\neg\varphi\}$, which is at most $-\frac{1}{2}\varepsilon_\emptyset < 0$ when B does not decide φ; for M_A the same sum is $\mathbb{U}\{\omega \in \mathrm{Cons} : A_\omega$ decides neither$\} \geq 0$.*

Proof of Proposition A.22. (a) Let $B \cup A \cup D^{\pm}$ be consistent; otherwise every term is 0. If $B \cup A \vdash \varphi^{b^*}$, then $D + (\varphi, b^*)$ keeps the value (it is consistent with A and has factor 1), and $D + (\varphi, 1 - b^*)$ is inconsistent with A. If A decides neither φ nor $\neg\varphi$, each extension has factor at most $\frac{1}{2}$. Integrating over Cons preserves the inequality, and dividing by $M_A^{1/2}(\emptyset) = 1 - \mathrm{Inc}_0 > 0$ gives a weak semimeasure with value 1 at $\emptyset$, to which Proposition 3.3 of the companion paper applies. (b) For a consistent A that proves ψ_D, $\nu_A^{1/2}(D) = 1$. (c) For each consistent A the new datum keeps consistency (it is entailed) and multiplies the value by 1 or $\frac{1}{2}$; for $A = \emptyset$ the value drops from $\frac{1}{2}$ to $\frac{1}{4}$. (d) Per consistent theory: if it proves $p \wedge q$, both values are 1; if it proves exactly one of p, q and is consistent with the other, both are $\frac{1}{2}$; if it proves neither and is consistent with $p \wedge q$, they are $\frac{1}{2}$ and $\frac{1}{4}$; otherwise both are 0. Integrate; the empty theory contributes at least $\varepsilon_\emptyset/4$ to the difference. (e) The Möbius sum on the inputs $\varphi, \neg\varphi$ at

$\emptyset$ is $\nu(\emptyset) - \sum_b \nu(\{(\varphi, b)\}) - \sum_b \nu(\{(\neg\varphi, b)\}) + \sum_{b,b'} \nu(\{(\varphi, b), (\neg\varphi, b')\})$, which we compute per consistent theory A for $\nu = \nu_A^{1/2}$. If A decides φ, say $B \cup A \vdash \varphi^{b^*}$, the singletons consistent with A are (φ, b^*) and $(\neg\varphi, 1 - b^*)$, both proved (value 1), and the only pair consistent with A is $\{(\varphi, b^*), (\neg\varphi, 1 - b^*)\}$ (value 1); the sum is $1 - 1 - 1 + 1 = 0$. If A decides neither, all four singletons are consistent and unproved (value $\frac{1}{2}$ each); of the four pairs, $\{(\varphi, 1), (\neg\varphi, 0)\}$ and $\{(\varphi, 0), (\neg\varphi, 1)\}$ are consistent and unproved (value $\frac{1}{4}$ each), and the other two are inconsistent; the sum is $1 - 2 + \frac{1}{2} = -\frac{1}{2}$. Integrate over Cons; the empty-theory tapes decide nothing that B does not decide. For M_A the same computation gives 0 for a deciding theory and $1-0-0+0 = 1$ for an undeciding one, since an undeciding theory proves no singleton datum on these inputs; so its sum is the undecided mass. By Proposition A.2 of the companion paper, a weak ν with $\nu(\emptyset) = 1$ is a containment functional iff all its Möbius sums are nonnegative. □

Proposition A.23 (The consistency-requiring predictor is a mixture)**.** *For every D with $M_A^{\mathrm{nc}}(D) > 0$ and every φ,*

$$M_A^{\mathrm{nc}}(D)\,\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) = \mathbb{U}(E_D \cap O_\varphi) + \mathbb{U}\{\omega : H_\omega \text{ consistent},\ B \cup A_\omega \nvdash \psi_D,\ H_\omega \vdash \varphi\},$$

and $\mathbb{U}(E_D \cap O_\varphi) = M_A(D)\mathrm{Bel}_D(\varphi)$ when $M_A(D) > 0$. So the consistency-requiring predictor is the mixture, with weight $M_A(D)/M_A^{\mathrm{nc}}(D)$, of the proof-requiring predictor and a predictor that uses weak theories together with the data.

Proof. Split the event $\{H_\omega \text{ consistent},\ H_\omega \vdash \varphi\}$ by whether $B \cup A_\omega \vdash \psi_D$. If it does, H_ω is equivalent to $B \cup A_\omega$: the consistency of H_ω is that of $B \cup A_\omega$, and $H_\omega \vdash \varphi$ iff $B \cup A_\omega \vdash \varphi$. So that part of the event is $E_D \cap O_\varphi$. □

The weak component contains the empty theory, of mass at least $\varepsilon_\emptyset$, which predicts from the data alone; it is the source of Proposition A.21(a).

Proposition A.24 (How much the graded weight can exceed M_A)**.** *(a) For every consistent D: $0 \le \log\left(M_A^{1/2}(D)/M_A(D)\right) \le K(\mathrm{dom}\, D) + |D| + c$. (b) Let $m \ge 3$ and $k \ge 1$, let $a_{j,i}$ ($j \le k$, $i < m$) be fresh atoms, and put $\chi_i^{(j)} := \neg a_{j,1} \wedge \cdots \wedge \neg a_{j,i-1} \wedge a_{j,i}$ for $1 \le i < m$ and $\chi_m^{(j)} := \neg a_{j,1} \wedge \cdots \wedge \neg a_{j,m-1}$. For $\vec{\imath} \in [m]^k$ let $D_{\vec{\imath}} := \{(\chi_{i_j}^{(j)}, 1) : j \le k\}$. Then some $\vec{\imath}$ has*

$$\log \frac{M_A^{1/2}(D_{\vec{\imath}})}{M_A(D_{\vec{\imath}})} \ \ge\ k \log \frac{m}{2} + \log \varepsilon_\emptyset, \qquad |D_{\vec{\imath}}| = k, \qquad K(\mathrm{dom}\, D_{\vec{\imath}}) \le k \log m + K(k, m) + c.$$

For $m = 3$ the gap is $k \log \frac{3}{2} - c$, linear in $|D|$; for $k = 1$ it is $\log m - 1 - c$ on a single datum. The lower bound of (b) is within $2|D| + K(k, m) + c$ of the upper bound of (a). The source is the per-datum grading of Proposition A.22(d): a datum whose content is a long conjunction costs one bit under the graded weight and its full content under M_A; m pairwise incompatible single data, each worth $\frac{1}{2}$ to the empty theory, have total graded mass $m/2 > 1$.

Proof. (a) The lower bound is Proposition A.22(b). For the upper bound, $M_A^{1/2} \le 1$, and $M_A(D) \ge 2^{-K(\psi_D) - c}$ by the one-axiom theory $\{\psi_D\}$, with $K(\psi_D) \le K(\mathrm{dom}\, D) + |D| + c$, since the labels are $|D|$ bits given $\mathrm{dom}\, D$. (b) Each $D_{\vec{\imath}}$ is consistent, since each $\chi_i^{(j)}$ is a consistent conjunction of literals and the blocks use disjoint atoms (L1), and B proves none of its data (L2 with $\Lambda = \emptyset$). The empty theory gives $\nu_\emptyset^{1/2}(D_{\vec{\imath}}) = 2^{-k}$, so $M_A^{1/2}(D_{\vec{\imath}}) \ge \varepsilon_\emptyset 2^{-k}$. Two different $\vec{\imath}$ differ in some block, where their sentences are incompatible; so the m^k conjunctions $\psi_{D_{\vec{\imath}}}$ are

pairwise incompatible, $\sum_{\vec{\imath}} M_A(D_{\vec{\imath}}) \le M_A(\emptyset) \le 1$, and some $\vec{\imath}$ has $M_A(D_{\vec{\imath}}) \le m^{-k}$. Its gap is at least $\log(\varepsilon_\emptyset 2^{-k} m^k)$. Finally $\operatorname{dom} D_{\vec{\imath}}$ is computable from (k,m) and the position of $\vec{\imath}$ among the m^k values. □

A.8 Limits and identification

The following lemma is the averaging argument associated with de Leeuw, Moore, Shannon and Shapiro (cf. de Leeuw et al., 1956), in the form we need. It is used again in **??**.

Lemma A.25 (Majority vote)**.** *Let Z be a decidable set of sentences, M an oracle machine that on tape ω and input $\sigma \in Z$ outputs $\mathsf{M}^\omega(\sigma) \in \{0,1\}$ or diverges, $S \subseteq Z$, $\kappa : S \to \{0,1\}$, and $\mathcal{G}$ a Borel set of tapes with $\mathbb{U}(\mathcal{G}) > 0$ such that for every $\omega \in \mathcal{G}$, $\mathsf{M}^\omega(\sigma)$ is defined for all $\sigma \in Z$ and equals $\kappa(\sigma)$ for $\sigma \in S$. Then some total computable $h : Z \to \{0,1\}$ extends κ.*

Proof. By outer regularity there is an open $U \supseteq \mathcal{G}$ with $\mathbb{U}(U) < \frac{4}{3}\mathbb{U}(\mathcal{G})$; writing U as a disjoint union of cylinders, some cylinder $[u]$ has $\mathbb{U}(\mathcal{G} \cap [u]) > \frac{3}{4}\, 2^{-|u|}$. For $b \in \{0,1\}$ let $Q_b(\sigma) := \{\omega \in [u] : \mathsf{M}^\omega(\sigma) = b\}$. These classes are effectively open uniformly in σ (a halting computation reads a finite prefix) and disjoint for $b = 0, 1$, so their measures are left-c.e., with computable approximations from below. Let $h(\sigma)$ be the b with the larger approximated measure at the first stage at which the two approximations sum to more than $\frac{1}{2}\, 2^{-|u|}$. That stage exists for every $\sigma \in Z$, because $\mathbb{U}(Q_0(\sigma) \cup Q_1(\sigma)) \ge \mathbb{U}(\mathcal{G} \cap [u]) > \frac{3}{4}\, 2^{-|u|}$. For $\sigma \in S$ the wrong class $Q_{1-\kappa(\sigma)}(\sigma)$ lies in $[u] \setminus \mathcal{G}$, of measure less than $\frac{1}{4}\, 2^{-|u|}$; so at that stage the approximation of the right class exceeds $\frac{1}{4}\, 2^{-|u|}$, more than that of the wrong one. □

Throughout the rest of this subsection D_∞ is a consistent infinite stream, $E_n := E_{D_n}$ (a decreasing sequence), $E_\infty := \bigcap_n E_n = \{\omega : B \cup A_\omega \text{ consistent, } \mathrm{Cn}(A_\omega) \supseteq D_\infty^{\pm}\}$, $a_n := \mathbb{U}(E_n)$, $a := \mathbb{U}(E_\infty)$ and $P_n := \mathbb{U}(\cdot \mid E_n)$.

Proposition A.26 (The limit posterior)**.** *(a) If $a > 0$, then $\sup_{\mathcal{X}} |P_n(\mathcal{X}) - P_\infty(\mathcal{X})| \le (a_n - a)/a_n \to 0$, the supremum over Borel sets of tapes, with $P_\infty := \mathbb{U}(\cdot \mid E_\infty)$. Hence $\sup_\chi |\mathrm{Bel}_{D_n}(\chi) - \mathrm{Bel}_\infty(\chi)| \to 0$ with $\mathrm{Bel}_\infty(\chi) := P_\infty(O_\chi)$, likewise for* Dis *and* Indep*, and the cumulative loss in the semimeasure convention converges to $-\log(a/M_A(\emptyset))$. (b) If $a = 0$, then $M_A(D_n) \to 0$ and the cumulative loss tends to ∞. (c) $a \ge \mathbb{U}_A(\mathrm{Ext}T) \ge 2^{-K_{\equiv}(T)-c}$ whenever some r.e. T with $B \cup T$ consistent proves every datum of D_∞. (d) For every set Γ of sentences with $\mathrm{Cn}(\Gamma) = \mathrm{Cn}(D_\infty^{\pm})$, complete or not, $E_\infty = \{\omega : B \cup A_\omega$ consistent, $\mathrm{Cn}(A_\omega) \supseteq \mathrm{Cn}(\Gamma)\}$; so, if $a > 0$, P_∞ is the prior restricted to the tapes whose theory is a consistent extension of Γ.*

Proof. (a) For a Borel $\mathcal{X}$ let $x := \mathbb{U}(\mathcal{X} \cap E_\infty) \le a$ and $y_n := \mathbb{U}(\mathcal{X} \cap (E_n \setminus E_\infty)) \le a_n - a$. Then

$$\Big|\frac{x + y_n}{a_n} - \frac{x}{a}\Big| = \frac{|y_n a - x(a_n - a)|}{a\, a_n} \le \frac{\max(y_n a,\, x(a_n - a))}{a\, a_n} \le \frac{a_n - a}{a_n},$$

and $a_n \downarrow a$ by continuity from above. The loss is $L_A(D_n) - L_A(\emptyset) = -\log(a_n/M_A(\emptyset))$ by **??**(d). (b) Continuity from above. (c) As in the proof of **??**. (d) A theory has $D_\infty^{\pm}$ among its consequences iff it contains $\mathrm{Cn}(D_\infty^{\pm}) = \mathrm{Cn}(\Gamma)$. □

Proposition A.27 (Limits)**.** *Let T^* be consistent and r.e. (a) If $B \cup T^*$ is complete (hence decidable) and $\mathrm{Cn}(D_\infty^{\pm}) = \mathrm{Cn}(T^*)$, then $P_n\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\} \to 1$, $\sup_\chi |\mathrm{Bel}_{D_n}(\chi) - \mathbf{1}[T^* \vdash \chi]| \to 0$, $\sup_\chi \mathrm{Indep}_{D_n}(\chi) \to 0$, and the cumulative loss converges to $-\log(\mathbb{U}_A[T^*]/M_A(\emptyset)) \le K_{\equiv}(T^*) + c$. If T^* is axiomatised by one sentence τ and $(\tau, 1) \in D_n$, the first probability is* 1 *from*

then on. (b) If $B \cup T^*$ is incomplete and $\mathrm{Cn}(D^{\pm}_{\infty}) = \mathrm{Cn}(T^*)$, then P_∞ is the prior restricted to the consistent extensions of T^*, and for every ψ independent of $B \cup T^*$:

$$\lim_n \mathrm{Bel}_{D_n}(\psi) = \frac{\mathbb{U}_A(\mathrm{Ext}(T^* + \psi))}{\mathbb{U}_A(\mathrm{Ext}T^*)} \in (0,1), \qquad \lim_n \mathrm{Dis}_{D_n}(\psi) = \frac{\mathbb{U}_A(\mathrm{Ext}(T^* + \neg\psi))}{\mathbb{U}_A(\mathrm{Ext}T^*)} \in (0,1),$$

$$\lim_n \mathrm{Indep}_{D_n}(\psi) \geq \frac{\mathbb{U}_A[T^*]}{\mathbb{U}_A(\mathrm{Ext}T^*)} > 0.$$

The most probable limit class can be a strict extension: in the literal setting, with fresh atoms α_i on a computable sequence of distinct words, $U := \{\alpha_i : i \in \mathbb{N}\}$ and $T_j := U \setminus \{\alpha_j\}$, for all but finitely many j the limit posterior on data presenting T_j gives the class of U more probability than the class of T_j; a most probable class exists, and it strictly extends T_j. (c) If every sentence appears with its T^*-trichotomy label, then $\mathbb{U}(\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\} \mid E^3_{D_n}) \to 1$.

Proof of Proposition A.27. (a) A complete consistent theory has no proper consistent extension, so $E_\infty = \{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\}$ by Proposition A.26(d), and this event has positive measure by **??**(i). On it the trichotomy is $\mathbf{1}[T^* \vdash \chi]$ for all χ at once. Apply Proposition A.26(a); the limit loss is $-\log(\mathbb{U}_A[T^*]/M_A(\emptyset)) \leq K_{\equiv}(T^*) + c$ by **??**(i). If T^* is axiomatised by τ and $(\tau, 1) \in D_n$, then every consistent theory proving ψ_{D_n} contains $\mathrm{Cn}(T^*)$, so $E_n = E_\infty$.

(b) E_∞ is as stated by Proposition A.26(d), and $\lim_n \mathrm{Bel}_{D_n}(\psi) = P_\infty(O_\psi)$ by Proposition A.26(a), likewise for Dis and Indep. Since $T^* + \psi$, $T^* + \neg\psi$ and T^* are consistent r.e. theories, all three ratios are positive by **??**(i); each of the first two is less than 1 because the other is positive and the two events are disjoint; and the tapes with $\mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)$ lie in the independent part. For the example: $T_j \nvdash \alpha_j$ by (L2) of Lemma A.14, while $T_{j'} \vdash \alpha_j$ for $j' \neq j$; so the classes $[T_j]$ are pairwise distinct and their events disjoint, and $\sum_j \mathbb{U}_A[T_j] \leq 1$, so $\mathbb{U}_A[T_j] \to 0$. U is r.e. with an index of complexity $O(1)$ and is consistent by (L1), so $\mathbb{U}_A[U] \geq 2^{-c}$. The limit posterior odds of $[U]$ against $[T_j]$ are $\mathbb{U}_A[U]/\mathbb{U}_A[T_j]$, which exceed 1 for all but finitely many j. *A most probable class exists:* under P_∞ the classes partition E_∞, so only finitely many classes have limit probability above the positive number $\mathbb{U}_A[U]/\mathbb{U}_A(\mathrm{Ext}T_j)$, which is the limit probability of $[U]$; a maximiser exists among them, lies in $\mathrm{Ext}(T_j)$, and is not $[T_j]$, which $[U]$ beats; so it is a strict extension of T_j.

(c) A consistent theory that proves every theorem of T^* and decides no sentence independent of $B \cup T^*$ has the same consequences as T^*: a theorem $\sigma \notin \mathrm{Cn}(T^*)$ of it would be refutable from T^* (then the theory is inconsistent) or independent of T^* (then σ is labelled ι and decided). So the three-valued analogue of E_∞ is $\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\}$, of positive measure by **??**(i), and the proof of Proposition A.26(a) applies to M^3_A unchanged. □

Since $\sum_j \mathbb{U}_A[T_j] \leq 1$, the same disjointness argument bounds $\mathbb{U}_A[T]$ for all but finitely many members of any infinite family of pairwise inequivalent theories with a common simple consistent extension. Two-valued data cannot do better than Proposition A.27(b), whatever the learner. Call a sequence (σ_t) of sentences a *text* for T if $\mathrm{Cn}(\{\sigma_t : t \geq 1\}) = \mathrm{Cn}(T)$; it presents T by the two-valued data $(\sigma_t, 1)$, a datum $(\varphi, 0)$ being the datum $(\neg\varphi, 1)$. A function F from finite sequences of sentences to deductive classes *identifies* $[T]$ on a text if its values on the prefixes of the text are eventually $[T]$.

Proposition A.28 (Text against informant)**.** *In the literal setting, let α_i be fresh atoms on a computable sequence of distinct words, $\Lambda_k := \{\alpha_i : i < k\}$ and $\Lambda_\infty := \{\alpha_i : i \in \mathbb{N}\}$. No function F from finite sequences of sentences to deductive classes, computable or not, identifies on every text every member of $\{[\Lambda_k] : k \geq 1\} \cup \{[\Lambda_\infty]\}$. In particular no learner identifies all consistent*

r.e. theories from two-valued data, while three-valued data identify every consistent r.e. T^* in the limit posterior (Proposition A.27(c)), non-computably.

Proof. Build a text for Λ_∞ in stages. Stage $k \geq 1$ appends α_{k-1} repeatedly until F outputs $[\Lambda_k]$. This happens, because the sequence built so far, continued by α_{k-1} forever, is a text for Λ_k, on which F eventually outputs $[\Lambda_k]$. The resulting infinite sequence contains every α_i, so it is a text for Λ_∞; and F outputs $[\Lambda_k] \neq [\Lambda_\infty]$ for infinitely many k, since $\Lambda_k \nvdash \alpha_k$ by (L2) of Lemma A.14. □

This is Gold's distinction between learning from text and from an informant (cf. Gold, 1967): a class that contains all finite languages and one infinite language is not identifiable from text, by the same locking argument. So the failure of two-valued SAI to identify incomplete truths is not an artefact of its prior.

Proposition A.29 (Computable structures; the basic ω-gap)**.** *Let $\mathfrak{M}$ be a computable structure in which every element is the value of a closed term, let* $\mathrm{Diag}(\mathfrak{M})$ *be its atomic diagram (the closed atomic sentences and negations of closed atomic sentences true in $\mathfrak{M}$; a decidable set), and let the data list* $\mathrm{Diag}(\mathfrak{M})$ *in a computable order, labelled true. (a) If B is true in $\mathfrak{M}$, then $L_A(D_n) \leq K_{\equiv}(\mathrm{Diag}\,\mathfrak{M}) + c \leq K(\mathfrak{M}) + c$, with $K(\mathfrak{M})$ the complexity of an index of $\mathfrak{M}$. (b) P_∞ is the prior restricted to the consistent extensions of* $\mathrm{Diag}(\mathfrak{M})$*. For $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, $\mathfrak{M} = (\mathbb{N}, 0, S, +, \cdot, <)$, $B = \emptyset$ (or any B true both in $\mathbb{N}$ and in the model $\mathbb{N} \sqcup \{\star\}$ of the proof) and $\vartheta := \forall x\,(x + 0 = x)$, the limits $\lim_n \mathrm{Bel}_{D_n}(\vartheta)$ and $\lim_n \mathrm{Dis}_{D_n}(\vartheta)$ both lie in $(0, 1)$: instances never settle the universal sentence. (If instead $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, then $B \vdash \vartheta$ and $\lim_n \mathrm{Bel}_{D_n}(\vartheta) = 1$.) (c) If* $\mathrm{Th}(\mathfrak{M})$ *is decidable and the data present it, Proposition A.27(a) applies.*

Proof. (a) **??** with $T = \mathrm{Diag}(\mathfrak{M})$, which is consistent with B, proves every datum, and has an index computable from one of $\mathfrak{M}$. (b) Proposition A.27(b) with $\psi = \vartheta$; it needs $\mathrm{Diag}(\mathbb{N}) \cup \{\vartheta\}$ and $\mathrm{Diag}(\mathbb{N}) \cup \{\neg\vartheta\}$ to be consistent with B. The first holds in $\mathbb{N}$. For the second take $\mathbb{N} \sqcup \{\star\}$ with $S(\star) := \star$, $\star + y := 0$ for every y (including $y = \star$), $y + \star := \star$ for $y \in \mathbb{N}$, $\star \cdot y = y \cdot \star := \star$ for every y, and $\star$ incomparable under $<$ with everything. Closed terms take values in $\mathbb{N}$, which is closed under the operations, so the diagram holds; and $\star + 0 = 0 \neq \star$. □

This is the one-tape form of the coin-free ω-gap of Theorem 3.9 and Proposition B.8 of the earlier report; **??** studies its anatomy.

Proposition A.30 (Complete truths that are not decidable)**.** *(a) If* $\mathrm{Cn}(D^{\pm}_{\infty})$ *is complete and not decidable (for example $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, $B \subseteq \mathrm{Th}(\mathbb{N})$ decidable, and every sentence labelled by its truth in $\mathbb{N}$), then $a = 0$: $M_A(D_n) \to 0$, and the cumulative loss in the semimeasure convention is unbounded. (b) If (σ_t) is a computable enumeration of all sentences and D_n labels $\sigma_1, \ldots, \sigma_n$ by a complete consistent theory, with label vector $b_{1:n}$, then $K(b_{1:n} \mid n) - c \leq L_A(D_n) \leq n + K(n) + c$.*

Proof. (a) On E_∞ the theory is consistent and contains the complete set $\mathrm{Cn}(D^{\pm}_{\infty})$, so it has exactly these consequences. Let $\mathsf{M}^{\omega}(\sigma)$ be 1 or 0 according as σ or $\neg\sigma$ is found first to be provable from $B \cup A_\omega$. On E_∞ it is total and decides $\mathrm{Cn}(D^{\pm}_{\infty})$. If $a > 0$, Lemma A.25, with $Z = S := \mathrm{Sent}$, $\mathcal{G} = E_\infty$ and κ the characteristic function of $\mathrm{Cn}(D^{\pm}_{\infty})$, gives a decision procedure for it, a contradiction. ($\mathrm{Th}(\mathbb{N})$ is not decidable, for instance by the inseparability step in the proof of Proposition A.31.) (b) The lower bound is **??**(b), since $\mathrm{dom}\, D_n$ is computable from n. For the upper bound, $K_A(D_n) \leq K(\psi_{D_n}) + c \leq K(n) + n + c'$, since given n the labels are n bits; then use $L_A \leq K_A + c$ (**??**). □

The coin-free analogues are Proposition 7.11 and Lemma E.11 of the earlier report; **??** gives the growth rate on Th(ℕ).

For the next proposition, F is a function machine in the sense of Definition 3.6 of the companion paper on the inputs $\mathrm{Sent}_{\mathcal{L}_{\mathrm{ar}}}$ with labels $\{0,1\}$, and f^F_ω its random partial labeller. We use two standard facts about Q (see Boolos et al., 2007): it strongly represents every computable set, and it satisfies the diagonal lemma.

Proposition A.31 (Random theories above Q are almost surely incomplete)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$. (a) For every theory machine G, the set of tapes ω such that $B \cup A^G_\omega$ is consistent, proves every axiom of Q, and decides every $\mathcal{L}_{\mathrm{ar}}$-sentence has measure 0. In particular, if $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, the consistent random theory of M_A is almost surely incomplete, and every posterior gives probability 0 to complete theories. (b) For every function machine F as above: $\mathbb{U}\{\omega : f^F_\omega$ is total, labels every Q-provable sentence 1 and every Q-refutable sentence $0\} = 0$; likewise with PA, or any consistent r.e. extension of Q, in place of Q.*

Proof. Inseparability. No total computable $h : \mathrm{Sent}_{\mathcal{L}_{\mathrm{ar}}} \to \{0,1\}$ is 1 on the theorems of Q and 0 on its refutable sentences. Otherwise the set $\{\ulcorner\sigma\urcorner : h(\sigma) = 1\}$ is strongly represented in Q by some formula $\rho_h(x)$, and the diagonal lemma gives σ with $\mathsf{Q} \vdash \sigma \leftrightarrow \neg\rho_h(\ulcorner\sigma\urcorner)$. If $h(\sigma) = 1$, then $\mathsf{Q} \vdash \rho_h(\ulcorner\sigma\urcorner)$, so $\mathsf{Q} \vdash \neg\sigma$, and h is 1 on a refutable sentence. If $h(\sigma) = 0$, then $\mathsf{Q} \vdash \neg\rho_h(\ulcorner\sigma\urcorner)$, so $\mathsf{Q} \vdash \sigma$, and h is 0 on a theorem. A separator for a consistent r.e. $T \supseteq \mathsf{Q}$ separates the pair of Q too.

(a) On the event, $\mathsf{M}^\omega(\sigma)$, the first of $1, 0$ found according as σ or $\neg\sigma$ is provable from $B \cup A^G_\omega$, is total on $\mathcal{L}_{\mathrm{ar}}$-sentences and separates the theorems of Q from its refutable sentences: a consistent theory that proves Q proves every theorem of Q and refutes every refutable sentence. If the event had positive measure, Lemma A.25 (with $Z := \mathrm{Sent}_{\mathcal{L}_{\mathrm{ar}}}$ and S the sentences Q decides) would give a computable separator. (b) The same, with $\mathsf{M}^\omega(\sigma) := f^F_\omega(\sigma)$, computed by running F on σ with tape ω. □

The proposition is a new application of the fact that separating sets of effectively inseparable pairs have measure zero (for effective inseparability, cf. Smullyan, 1961); Lemma A.25 makes the proof self-contained. Part (b) says that a random tape does not help a labeller to get every Q-decided sentence right.

Corollary A.32 (Local defects above Q)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and let G be a theory machine with $\nu_G(\emptyset) > 0$ whose random theory proves Q almost surely on Cons^G (for instance, any G when $\mathsf{Q} \subseteq \mathrm{Cn}(B)$). Then ν_G has positive local defect at $D = \emptyset$ at some $\mathcal{L}_{\mathrm{ar}}$-sentence φ: $\nu_G(\{(\varphi,1)\}) + \nu_G(\{(\varphi,0)\}) < \nu_G(\emptyset)$. In particular, above Q there is no theory machine whose law is "a theory plus a fair coin on every sentence it leaves undecided" with zero local defect everywhere.*

Proof. By **??**(b) at $D = \emptyset$, the defect at φ is $\nu^3_G(\{(\varphi,\iota)\})$, the measure of the set of $\omega \in \mathrm{Cons}^G$ such that $B \cup A^G_\omega$ decides neither φ nor $\neg\varphi$. If it were 0 for every $\mathcal{L}_{\mathrm{ar}}$-sentence, then, by a countable union, almost every $\omega \in \mathrm{Cons}^G$ would have a theory that decides every $\mathcal{L}_{\mathrm{ar}}$-sentence; with the hypothesis, the event of Proposition A.31(a) would have measure $\mathbb{U}(\mathrm{Cons}^G) = \nu_G(\emptyset) > 0$, a contradiction. □

A.9 Order dependence and the Kolmogorov predictor

In the normalised convention (Definition 1.5) each conditional of M_A is divided by $\mathrm{Bel}_D(\varphi) + \mathrm{Dis}_D(\varphi)$, so the loss of a reveal sequence is no longer a function of the final data set.

Theorem A.33 (Order dependence in the normalised convention)**.** *(a) Let a, χ be sentences such that $B \cup \{a\}$ is consistent, and $B \cup \{\neg a\}$ is consistent and decides neither χ nor $\neg\chi$. Let $D := \{(a,1), (a \vee \chi, 1)\}$, and for $x, y \in Y_3$ let $p_{xy} := \mathbb{U}\{\omega \in \mathrm{Cons} : \mathrm{tri}_\omega(a) = x,\ \mathrm{tri}_\omega(a \vee \chi) = y\}$. Then $p_{10} = p_{1\iota} = p_{\iota 0} = 0$, and the normalised probabilities q_1 and q_2 of D in the orders $(a, a \vee \chi)$ and $(a \vee \chi, a)$ satisfy*

$$\frac{q_2}{q_1} = 1 + \frac{(p_{11} + p_{01})\, p_{0\iota} + p_{\iota 1}\,(p_{00} + p_{0\iota})}{(p_{11} + p_{01} + p_{\iota 1} + p_{00})\,(p_{11} + p_{01})} \ > \ 1.$$

So the normalised cumulative loss of M_A on D is strictly smaller in the order $(a \vee \chi, a)$, while the loss in the semimeasure convention is the same in both orders. (b) Let a_i, a_i' $(i \geq 1)$ be fresh atoms on pairwise distinct words. For $y \in \{0,1\}^n$ put $\alpha_i := a_i^{y_i}$, $\beta_i := \alpha_i \vee a_i'$ and $D_n(y) := \{(\alpha_i, 1), (\beta_i, 1) : i \leq n\}$, and compare the orders $(\alpha_1, \beta_1, \alpha_2, \beta_2, \dots)$ and $(\beta_1, \alpha_1, \beta_2, \alpha_2, \dots)$. For every y the normalised loss of the first order exceeds that of the second by at least $\sum_{i \leq n} \log(1 + v_i(y)) \geq 0$, where $v_i(y)$ is the posterior probability, after the first $i-1$ pairs, that the random theory refutes α_i and does not decide a_i'. The average of this difference over $y \in \{0,1\}^n$ is at least $2^{-c_1 - 1} n$, where c_1 is the length of a V-program for the machine $G_{\mathrm{ca}} := \{(w, a_j^{w_j}) : |w| = j\}$ (fair coins on the atoms a_j). So for every n some data set of $2n$ data has two reveal orders whose normalised losses differ by at least $2^{-c_1 - 1} n$ bits.

Proof. (a) If the random theory proves a, it proves $a \vee \chi$; if it refutes $a \vee \chi$, it refutes a. This gives the three zero cells. In the order $(a, a \vee \chi)$ the first normalised factor is $\mathrm{Bel}_\emptyset(a)/(\mathrm{Bel}_\emptyset(a) + \mathrm{Dis}_\emptyset(a)) = p_{11}/(p_{11} + p_{01} + p_{00} + p_{0\iota})$, and the second is 1, since $a \vee \chi$ is then entailed (**??**(c)). In the order $(a \vee \chi, a)$ the first factor is $(p_{11} + p_{01} + p_{\iota 1})/(p_{11} + p_{01} + p_{\iota 1} + p_{00})$; given $(a \vee \chi, 1)$ the posterior is the prior restricted to $\mathrm{tri}_\omega(a \vee \chi) = 1$, on which $\mathrm{Bel}(a) \propto p_{11}$ and $\mathrm{Dis}(a) \propto p_{01}$, so the second factor is $p_{11}/(p_{11} + p_{01})$. (The normalising constant $1 - \mathrm{Inc}_0$ cancels in each factor.) Dividing,

$$\frac{q_2}{q_1} = \frac{(p_{11} + p_{01} + p_{\iota 1})(p_{11} + p_{01} + p_{00} + p_{0\iota})}{(p_{11} + p_{01} + p_{\iota 1} + p_{00})(p_{11} + p_{01})},$$

and the numerator minus the denominator is $(p_{11} + p_{01})p_{0\iota} + p_{\iota 1}(p_{00} + p_{0\iota})$. For strictness: $p_{11} > 0$ (the tapes beginning with a program for $\{a\}$), and $p_{0\iota} > 0$ (the tapes beginning with a program for $\{\neg a\}$, which are consistent, refute a, and do not decide $a \vee \chi$, because $B \cup \{\neg a\} \vdash a \vee \chi$ iff $B \cup \{\neg a\} \vdash \chi$, and $B \cup \{\neg a\} \vdash \neg(a \vee \chi)$ iff $B \cup \{\neg a\} \vdash \neg\chi$).

(b) After the first $i-1$ pairs both orders have revealed the same set $D_{i-1}(y)$, and the normalised conditionals depend only on the current set and the queried sentence. So the difference is the sum over i of $\log(q_2^{(i)}/q_1^{(i)})$, computed as in (a) under the posterior $P_{i-1} := \mathbb{U}(\cdot \mid E_{D_{i-1}(y)})$ in place of the prior restricted to Cons; the zero cells of (a) hold for every consistent theory. Write $m_1 := p_{11} + p_{01}$, $m_2 := p_{\iota 1}$, $m_3 := p_{00}$ and $m_4 := p_{0\iota}$ for the cell masses under P_{i-1}; they sum to at most 1, and $m_1 > 0$ (the tapes of G_{ca} that agree with y up to i). The ratio is $1 + [m_1 m_4 + m_2(m_3 + m_4)]/[(m_1 + m_2 + m_3)m_1] \geq 1 + m_4/(m_1 + m_2 + m_3) \geq 1 + m_4 = 1 + v_i(y)$. Next, $\log(1+v) \geq v$ for $v \in [0,1]$. The tapes $p\,\omega'$, with p the program for G_{ca}, $\omega'_{<i} = y_{<i}$ and $\omega'_i \neq y_i$, lie in $E_{D_{i-1}(y)}$ (their theory proves α_j, hence β_j, for $j < i$), refute α_i, and do not decide a_i' (by (L2) of Lemma A.14); they have measure $2^{-c_1 - i}$. So $v_i(y) \geq 2^{-c_1 - i}/M_A(D_{i-1}(y))$. Over uniform y, Jensen's inequality gives $\mathbb{E}[1/M_A(D_{i-1})] \geq 1/\mathbb{E}[M_A(D_{i-1})]$. Moreover $M_A(D_{i-1}(y)) = \mathrm{Bel}_0(\bigwedge_{j<i} \alpha_j)$, and these conjunctions are pairwise incompatible for distinct $y_{<i}$, so $\sum_{y_{<i}} M_A(D_{i-1}(y)) \leq 1$ and $\mathbb{E}[M_A(D_{i-1})] \leq 2^{-(i-1)}$. Hence $\mathbb{E}\, v_i \geq 2^{-c_1 - 1}$; sum over $i \leq n$. □

The mechanism is the failure of the two-point criterion of Proposition A.7 of the companion paper: whether the random theory decides $a \vee \chi$ depends on its answer to a, since a theory proving

a always decides $a \vee \chi$ and one refuting a need not. The witnesses use logical structure: in one order the second datum is entailed. On logically independent data the question stays open.

Question A.34. Is the normalised loss of M_A order dependent on logically independent data, for instance on positive literals of a fresh predicate? This is Question A.8 of the companion paper in another guise: on such data $M_A \asymp M_F$ (**??**), but constant factors do not control normalised conditionals.

The single shortest compatible theory can be a poor predictor where the mixture is good. Let $a_1, \dots, a_m$ be fresh atoms, and for $w \in \{0,1\}^m$ let $\mathrm{desc}(w) := \bigwedge_{i \le m} a_i^{w_i}$.

Theorem A.35 (The Kolmogorov predictor can be exponentially worse than Bayes)**.** *Fix any well-order $\prec$ of $\mathrm{dom}\, V$, and let the Kolmogorov predictor of Proposition A.11(b) use the $\prec$-first compatible program. There are a data set D of $2^m - 1$ sentences, all of the form $\neg\mathrm{desc}(v)$ labelled true, and a reveal order of D, such that: (i) the Kolmogorov predictor errs or abstains at every one of the $2^m - 1$ steps; (ii) $K_A(D) \le m + K(m) + c$, and the Bayes predictor of Proposition A.11(a) makes at most $K_A(D) + c \le m + K(m) + c'$ mistakes on the same sequence. If $\prec$ refines length, then $m \le K_A(D) \le m + K(m) + c$; so with $k := K_A(D)$ the number of errors or abstentions is at least $2^{k - K(m) - c} - 1 \ge 2^{k - 2\log k - c'} - 1$, while the Bayes predictor makes at most $k + c$ mistakes. The upper bound of Proposition A.11(b) is 2^k.*

Proof. For each w let p_w be the $\prec$-first $p \in \mathrm{dom}\, V$ with $B \cup \mathsf{W}_{V(p)}$ consistent and proving $\mathrm{desc}(w)$. It exists, since $\{\mathrm{desc}(w)\}$ is consistent (Lemma A.14) and $\prec$ is a well-order. A consistent theory proves at most one $\mathrm{desc}(w)$, so $w \mapsto p_w$ is injective. Let w^* be the w whose p_w is $\prec$-last among these 2^m programs. Starting from $D = \emptyset$, the adversary repeats the following as long as the $\prec$-first program p compatible with the current data does not prove $\mathrm{desc}(w^*)$.

- p is consistent and $B \vdash \bigvee_v \mathrm{desc}(v)$, so the set $\mathrm{Open}(p)$ of v with $B \cup \mathsf{W}_{V(p)} \cup \{\mathrm{desc}(v)\}$ consistent is nonempty. It contains some $v \neq w^*$: otherwise p proves $\neg\mathrm{desc}(v)$ for all $v \neq w^*$, hence $\mathrm{desc}(w^*)$.
- Present $(\neg\mathrm{desc}(v), 1)$ for such a v.

Then: (1) every presented sentence is true in the valuation w^*, so the data stay consistent and $\{\mathrm{desc}(w^*)\}$ proves them all. (2) p does not prove $\neg\mathrm{desc}(v)$, since v is open for p; so the predictor errs (if p proves $\mathrm{desc}(v)$) or abstains. (3) The sentence $\neg\mathrm{desc}(v)$ was not presented before, since p is compatible with the earlier data. (4) For $w \neq w^*$, p_w proves $\neg\mathrm{desc}(v)$ for every $v \neq w$, so it stays compatible until $\neg\mathrm{desc}(w)$ is presented. While some p_w with $w \neq w^*$ is compatible, the $\prec$-first compatible program is $\preceq p_w \prec p_{w^*}$, and, being consistent and $\prec$-before p_{w^*}, it does not prove $\mathrm{desc}(w^*)$; so the loop continues. By (3) the loop runs at most $2^m - 1$ times, and by (4) it does not stop before every $\neg\mathrm{desc}(w)$, $w \neq w^*$, has been presented. So it runs exactly $2^m - 1$ times, and by (2) every step is an error or an abstention.

(ii) $\bigwedge_{v \neq w^*} \neg\mathrm{desc}(v)$ is equivalent to $\mathrm{desc}(w^*)$, so $K_A(D) \le K(\mathrm{desc}(w^*)) + c \le m + K(m) + c'$. For the Bayes predictor, apply Proposition A.11(a) with $T := \mathsf{W}_{V(p)}$ for a shortest program p compatible with D: $K_{\equiv}(T) \le |p| + c = K_A(D) + c$, and T proves every datum.

If $\prec$ refines length, every program compatible with D proves $\mathrm{desc}(w^*)$, and the $\prec$-first of them is p_{w^*}, so $K_A(D) = |p_{w^*}|$. The 2^m distinct programs p_w all have length at most $|p_{w^*}|$, and Kraft's inequality gives $2^m \le 2^{|p_{w^*}|}$. Finally $K(m) \le 2\log m + c \le 2\log k + c$. □

The stream is not computable, since it depends on the programs p_w. The construction is logical: in SOI on strings the analogous decoys would be beaten by the constant labeller "true",

which is not available here because $\{\neg\mathrm{desc}(v) : v \in \{0,1\}^m\}$ is inconsistent. As in the conjunction ladder of Theorem 9.10 of the companion paper, it is the selection rule that fails, not the prior.

Proof of Proposition A.11(c). For each $w \neq 0^m$ fix a program $q_w \in \mathrm{dom}\, V$ for an index of $\{\mathrm{desc}(w)\}$. Let $\prec$ list the $2^m - 1$ programs q_w first, in any order, and then the rest of $\mathrm{dom}\, V$ by length; this is a well-order of type ω. A consistent theory proves at most one $\mathrm{desc}(w)$, so $p_w = q_w$ for $w \neq 0^m$ (notation of the proof of Theorem A.35), and p_{0^m} lies in the tail; hence $w^* = 0^m$. Theorem A.35, which holds for every well-order, gives $2^m - 1$ errors or abstentions on a data set D whose conjunction is equivalent to $\mathrm{desc}(0^m)$, so $K_A(D) \leq K(\mathrm{desc}(0^m)) + c \leq K(m) + c'$; and $K(m) \leq 2\log m + c''$. □

A.10 The generative variant

Definition A.36 (Generative law)**.** Take $B = \emptyset$, or include B among the axioms. Fix the likelihood L_1 of Definition 2.9 of the earlier report: a derivation grammar with positive rule probabilities, a full-support instantiation grammar, and admissible parameters. For a set A of axioms with an enumeration $(\chi_1, \chi_2, \dots)$, give χ_j the citation weight proportional to 2^{-j} (normalised; no axiom is cited if $A = \emptyset$), and let P_A be the resulting law on sentences. The data are unlabelled sentences $\sigma_1, \sigma_2, \dots$. In the one-tape form, split the tape into $\omega^{(0)}, \omega^{(1)}, \dots$ by a computable interleaving; the theory is $A_\omega := A^{G_U}_{\omega^{(0)}}$, enumerated in the order of emission, and the i-th datum σ_i is sampled from P_{A_ω} with the coins $\omega^{(i)}$. The universal law of unlabelled data streams is $\Xi(\sigma_{1:n}) := \int \prod_{i \leq n} P_{A_\omega}(\sigma_i)\, d\mathbb{U}(\omega)$, and the generative posterior ϱ_n after n data has density proportional to $\prod_{i \leq n} P_{A_\omega}(\sigma_i)$ with respect to $\mathbb{U}$.

Consider the *support condition*: $\mathrm{supp}\, P_A = \mathrm{Cn}(A)$ for every set of axioms A. Lemma 2.10(c) of the earlier report proves it for finite template theories; we use it here as a hypothesis.

Proposition A.37 (Generative updating, given the support condition)**.** *Assume the support condition. (a)* Reweighting. *On data $\sigma_1, \dots, \sigma_n$, a consistent A has positive likelihood iff $A \vdash \sigma_i$ for all i, which is proof-requiring compatibility with $D := \{(\sigma_i, 1) : i \leq n\}$. So on consistent theories the generative posterior is the proof-requiring posterior reweighted by $\prod_i P_A(\sigma_i) \leq 1$. Inconsistent theories have positive likelihood, since their support is every sentence. (b)* Question conditioning. *For a yes/no question $\mathcal{Q} = \{\varphi, \neg\varphi\}$ and a consistent A: $P_A(\mathcal{Q}) > 0$ iff A decides φ, and then $P_A(\varphi^b \mid \sigma \in \mathcal{Q}) = \mathbf{1}[A \vdash \varphi^b]$. Hence, on the consistent theories, the generative predictive probability that the next datum answers $\mathcal{Q}$ by φ^b, given that it answers $\mathcal{Q}$, is*

$$\frac{\int \varrho_n(d\omega)\, P_{A_\omega}(\mathcal{Q})\, \mathbf{1}[A_\omega \vdash \varphi^b]}{\int \varrho_n(d\omega)\, P_{A_\omega}(\mathcal{Q})},$$

the proof-requiring belief under a posterior reweighted by how likely each theory was to produce an answer to $\mathcal{Q}$; inconsistent theories contribute their own $P_A(\varphi^b \mid \sigma \in \mathcal{Q})$. The outcome "independent" gets probability 0.

Proof. (a) By the support condition, $P_A(\sigma_i) > 0$ iff $\sigma_i \in \mathrm{Cn}(A)$. (b) $\mathrm{supp}\, P_A = \mathrm{Cn}(A)$ contains at most one of $\varphi, \neg\varphi$, by consistency, and contains one of them iff A decides φ; then apply Bayes' rule. □

Inconsistent theories pay the size principle: by Lemma 2.21 of the earlier report, a theory with mass ε outside the support of the data law loses at least $\log \frac{1}{1-\varepsilon}$ bits per datum in expectation. The generative variant thus models which sentences get stated; the outcome "independent" disappears because the model presupposes that asked questions are answered.

Remark A.38 (Supports and identification). We expect the following two statements to hold, and give only proof sketches. (i) *Supports*: the support condition holds for every axiom set in which every axiom has positive citation weight. (ii) *Identification*: if $\sigma_1, \sigma_2, \ldots$ are independent draws from P_{T^*}, the law of the deterministic machine that emits a consistent r.e. T^*, then the generative posterior probability of $\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\}$ tends to 1 almost surely; by contrast, the proof-requiring posterior on two-valued data presenting T^* reaches only the prior restricted to the consistent extensions of T^* (Proposition A.27(b)).

Proof sketch. We do not write out the support condition for r.e. axiom sets, nor the measurability of $\omega \mapsto P_{A_\omega}$ that (ii) needs. For (i), the argument of Lemma 2.10(c) of the earlier report applies to such axiom sets: each derivation tree that cites finitely many axioms has positive probability. For (ii), $\operatorname{supp} P_{A_\omega} = \mathrm{Cn}(A_\omega)$ is almost surely a function of the infinite sample (the set of sentences that occur), so Doob's consistency theorem (Theorem 4.2 of the earlier report; Doob, 1949) gives posterior consistency for it at $\mathbb{U}$-almost every ω; the tapes whose theory has the law P_{T^*} form a set of positive measure, so the statement holds almost surely under P_{T^*}.

A.11 Priors that complete their random theory

We compare M_A with two processes that draw sentences or theories until the theory is complete. We define them explicitly and prove their properties for the processes as defined. Process 1 is Demski's construction (Demski, 2012) as it is usually described. Process 2 is, at the level of the published descriptions, the modification of Demski's prior due to Fallenstein, Demski and Garrabrant (Garrabrant, 2015), used in Garrabrant et al. (2016), which samples machines on random bits and reads their outputs as sets of sentences; its sampling stage is that of G_U. We use nothing about either construction beyond the definitions below.

- *Process 1 (sentence draws).* Let μ be a probability on Sent with $\mu(\varphi) > 0$ for every φ. Draw sentences independently from μ, and keep a drawn sentence iff it is consistent with B and the sentences kept so far. Let S_t be the set kept after t draws, $T_\infty := \mathrm{Cn}(\bigcup_t S_t)$ and $P_\mu(\varphi) := \Pr[\varphi \in T_\infty]$.
- *Process 2 (theory draws).* At each step t draw an independent tape $\omega^{(t)}$ and the theory $A_{\omega^{(t)}}$; keep it iff it is consistent together with B and what has been kept. Let T'_∞ be the deductive closure of everything kept. (In a variant, the sentences of a drawn theory are kept one by one, as long as they are consistent with what has been kept.)

Proposition A.39 (Completing processes against M_A)**.** *(a) T_∞ is consistent and complete with probability* 1*. Hence P_μ is a probability on the Lindenbaum algebra of B ($P_\mu(\varphi) + P_\mu(\neg\varphi) = 1$), and $\Pr[\mathrm{Cn}(T_\infty) = \mathrm{Cn}(T)] = 0$ for every incomplete T. (b) T'_∞ is consistent and complete with probability* 1*, also in the variant; so Process 2 also gives a coherent probability over completions, and gives every incomplete theory class probability* 0*. (c) Under M_A, by contrast: for every φ undecided by B, the prior independent mass $\mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\varphi) - \mathrm{Bel}_0(\neg\varphi)$ is at least $\varepsilon_\emptyset > 0$; every r.e. T with $B \cup T$ consistent, complete or not, has $\mathbb{U}_A[T] \geq 2^{-K_{\equiv}(T)-c} > 0$; and if $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, the random theory is almost surely incomplete (Proposition A.31). (d) For any prior concentrated on complete theories, such as those of (a) and (b), proof-requiring and consistency-requiring compatibility coincide, and the trichotomy has no independent outcome (**??**).*

Proof. (a) Fix φ. With probability 1 it is drawn at some step, since $\mu(\varphi) > 0$. At the first such step t, either $B \cup S_{t-1} \cup \{\varphi\}$ is consistent and φ is kept, or $B \cup S_{t-1} \vdash \neg\varphi$; either way T_∞ decides φ. There are countably many sentences. Every S_t is consistent with B, so their union

is, by compactness. (b) Fix φ. At each step the drawn theory is $\{\varphi\}$ with probability at least $2^{-K(\varphi)-c} > 0$ (the tapes that begin with a program for the machine emitting φ), independently across steps; so with probability 1 some step draws $\{\varphi\}$. At the first such step, either $\{\varphi\}$ is consistent with what has been kept, and is kept, or what has been kept proves $\neg\varphi$. Either way T'_∞ decides φ. Take the countable union; consistency follows by compactness as in (a). In the variant, the same step decides φ. (c) The tapes with $A_\omega = \emptyset$ have mass $\varepsilon_\emptyset$ and decide only what B decides; the rest is **??**(i) and Proposition A.31. (d) is **??**. □

Running either process needs a consistency test at every step: already for $B = \emptyset$, a language with a binary relation symbol and one draw, it asks whether a sentence is satisfiable, which is undecidable by Church's theorem in Kalmár's form (Church, 1936; Kalmár, 1937). Process 1 is computable relative to $\emptyset'$, since consistency of a finite set is Π^0_1. Process 2 needs a consistency test relative to the tapes, which is Π^0_1 in them. Neither is computable, since a single draw of Process 2 can output one sentence. A theory machine emits sentences unconditionally, and consistency enters M_A only once, as the event measured (Proposition A.5). So Process 2 and M_A share the sampling stage, and with it the program-length credit for infinite schemas and dominance over the laws of machines; that stage is not new here. They differ in completion (iterated draws to a complete theory against a single draw, whose theory is incomplete with positive probability), in what a prediction is (a coherent probability over completions against a belief function of provability with an independent outcome), and in what is conditioned on (for complete theories the two compatibility notions coincide; for M_A they differ, and the proof-requiring one carries the learning guarantees of **??**).

A.12 Domination of provability functions

Recall from **??** that a random consistent theory Θ has the provability function $\beta(\chi) = \Pr[\chi \in \Theta]$, and that **??** asks whether $\mathrm{Bel}_0 \geq 2^{-c_\beta}\beta$ whenever β is lsc. Here "consistent" means consistent with B.

Proposition A.40 (Partial domination)**.** *(a) For every lsc provability function β, every finite set S of pairwise B-incompatible sentences and every $\sigma \in S$:* $\mathrm{Bel}_0(\sigma) \geq 2^{-K(\beta)-K(S)-c}\beta(\sigma)$. *In particular* $M_A(D) \geq 2^{-K(\beta)-K(\mathrm{dom}\, D)-c}\beta(\psi_D)$. *(b) If Θ is the deductive closure of the output of a theory machine G whose output is almost surely consistent, then* $\mathrm{Bel}_0 \geq 2^{-K(G)}\beta$. *(c) In the literal setting,* **??** *restricted to random sets of literals of R with at most one literal per word (with a lower semicomputable containment functional on literal data, and domination required on literal data) is equivalent to Question 3.14 of the companion paper for labels $\{0,1\}$: whether its one-tape law M_F dominates every lower semicomputable law of a random partial function, up to a constant factor.*

Proof. (a) Θ contains at most one member of S, by consistency, so $\sum_{\sigma\in S}\beta(\sigma) \leq 1$, and β restricted to S is a semimeasure, lsc uniformly in S. The conditional coding theorem (Section 1.1) gives $K(\sigma \mid S) \leq -\log\beta(\sigma) + K(\beta) + c$. If $\beta(\sigma) > 0$, then σ is consistent with B, and the one-axiom theory $\{\sigma\}$ gives $\mathrm{Bel}_0(\sigma) \geq 2^{-K(\sigma)-c} \geq 2^{-K(S)-K(\sigma|S)-c'}$. For D, take S to be the label conjunctions on $\mathrm{dom}\, D$. (b) Then $\beta(\chi) = \nu_G(\{(\chi,1)\})$, and **??** gives $\mathrm{Bel}_0(\chi) = M_A(\{(\chi,1)\}) \geq 2^{-K(G)}\nu_G(\{(\chi,1)\})$. (c) A random set Λ of literals with at most one literal per word corresponds to the random partial function h_Λ with $h_\Lambda(x) = b$ iff $R(t_x)^b \in \Lambda$, and $\Pr[\Lambda \supseteq D^\pm] = \Pr[h_\Lambda \supseteq D^{\mathrm{str}}]$, with D^{str} the string data set of D; and $M_A(D) \asymp M_F(D^{\mathrm{str}})$ on literal data (**??**). □

So **??** is at least as hard as the binary case of Question 3.14 of the companion paper, and for general sentences it is open. The obstacle is the same: a lower semicomputable law of a random object need not be realisable by a machine that reads one tape. Note that Bel_0 itself is not lower semicomputable (**??**), while $\mathrm{Bel}_0 + \mathrm{Inc}_0$ is; and by Corollary A.32, a theory-machine law above Q always has a local defect somewhere, so the question is not trivial.

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